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Action-Angle Clustering of Stellar Streams in Merging Galaxies

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ABSTRACT

Stellar streams are sensitive tracers of the Galactic gravitational field, but action-clustering methods are formally grounded in time-independent integrable potentials, whereas the Milky Way is neither static nor isolated. We test radial-action clustering using simulated globular-cluster streams in live Milky Way-like halos undergoing mergers with different mass ratios and orbital configurations.

For each stellar cohort and snapshot, we define a theoretical spherical target potential whose force best matches the true acceleration field over the region occupied by the cohort. The corresponding instantaneous actions evolve because of both the residual force not represented by this target and changes in the cohort-specific target potential. This framework motivates observable phase-space criteria designed to select stream segments whose action clouds are likely to remain compact. We then fit each selected cohort with a spherical NFW potential by maximizing the Kullback–Leibler divergence (KLD) of its radial-action distribution.

At the distance-matched snapshot of the 1:10 merger, the selected cohorts provide a useful recovery of the inner-halo enclosed-mass profile. Across the simulations, stronger mergers produce earlier,larger and more rapidly varying mass biases, whereas the 1:100 merger remains comparatively stable. Decomposing the fitted-to-true mass bias shows that errors in recovering the spherical target generally dominate over the difference between that target and the true mass profile in the illustrated inner-halo case. Broader target-action clouds are associated with larger positive KLD excesses and a wider range of mass biases. Earlier cohorts can remain orderly enough for a mismatched trial potential to compress their actions artificially, whereas later, member-dependent disturbances are less readily mapped into a highly clustered action cloud. Finally, near-degenerate KLD maxima need not produce large mass errors: their effect depends on whether the preferred direction crosses enclosed-mass contours. This motivates a post-fit identifiability screen based on the mass range spanned by the near-maximum KLD region. After applying stricter velocity cuts, a bending comparison at snapshot 55 improves target-parameter recovery for two of three surviving cohorts, whereas it worsens the third. Thus, geometry can help individual fits, although the improvement is not general. Observable cohort selection can therefore improve action-clustering fits in evolving halos, but cannot guarantee unique potential recovery or an unbiased global mass profile.

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1. INTRODUCTION

Stellar streams form through the tidal disruption of globular clusters and dwarf galaxies. Because their stars originate from dynamically cold progenitors and subsequently follow neighboring trajectories in the Galactic gravitational field, the debris forms thin, coherent structures in phase space. Their small internal phase-space dispersion allows them to retain information about the mass profile, shape, and dynamical evolution of the Milky Way halo.

One way to extract this information is through the distribution of stream stars in action space. In a time-independent integrable Hamiltonian, canonical actions label invariant orbital tori and remain constant along each orbit. Stars stripped from a dynamically cold progenitor are therefore expected to occupy neighboring regions of action space when their phase-space coordinates are evaluated in an appropriate integrable potential. This motivates potential-fitting methods that search for the potential model in which stream actions are maximally clustered. R. E. Sanderson et al. (2015) formulated this approach using the Kullback–Leibler divergence (KLD), and subsequent studies have applied and tested it using mock and observed stream samples (e.g., S. Reino et al. 2021, 2022). More generally, the construction and numerical accuracy of action coordinates depend on the integrability or approximate separability of the adopted potential (J. L. Sanders & J. Binney 2016).

Maximum clustering is not, however, identical to recovery of the correct potential. An incorrect trial potential can shift or compress an action distribution, and a short stream segment with limited orbital-phase coverage may not distinguish between competing models. Potential recovery therefore depends not only on the intrinsic coldness of the tracer but also on its coverage in orbital phase, radius, energy, and angular momentum. The orbital-phase dependence found by S. Reino et al. (2021), together with the improved constraints obtained by combining dynamically distinct streams, illustrates this identifiability problem.

A live merger creates an additional and more fundamental complication. The true Galactic Hamiltonian is time dependent and generally non-integrable, so a global set of conserved canonical actions for the full system need not exist. We address this problem without treating the true potential as integrable. For each material cohort C and each instant, we instead define a cohort-specific spherical target potential, $\Phi_{\mathrm{tar},C}(t)$, that best represents the true force over the region occupied by that cohort. Freezing this target potential defines an integrable action map for the instantaneous stellar phase-space points. As both the stars and their cohort-specific target potential evolve, the assigned target actions change through the residual force not reproduced by the target potential and through the change of the action map itself. Nearly common changes translate the action cloud without greatly altering its internal compactness, whereas member-dependent changes broaden or distort it.

This distinction motivates our observable cohort selection. We seek stream segments whose current morphology and kinematics indicate that their members probably experienced similar accumulated forcing and action-coordinate remapping. Smooth along-stream velocity profiles, unfolded radial phase-space branches, limited spatial extent, and separation from turning points cannot prove a semi-rigid target-action history, but they can reject material that is visibly folded, phase mixed, or kinematically irregular. The selection is therefore a physically motivated precondition for action clustering rather than a guarantee of an unbiased fit.

The Milky Way provides an important example of this non-equilibrium problem. Its halo is neither isolated nor exactly spherical, and it is perturbed by massive satellites and recent accretion events. In particular, the Large Magellanic Cloud (LMC) can generate a dynamical-friction wake and a large-scale response in the Milky Way halo (C. Vera-Ciro & A. Helmi 2013; N. Garavito-Camargo et al. 2019). Its infall also induces reflex motion between the inner Galaxy and the outer halo, which can bias dynamical inferences that assume a stationary Galactocentric frame (M. S. Petersen & J. Peñarrubia 2020). Observed and simulated stellar streams exhibit track–velocity misalignments, orbital deflections, and other perturbations associated with the LMC and the deformation of the Milky Way halo (e.g., D. Erkal et al. 2018; N. Shipp et al. 2019; R. A. N. Brooks et al. 2025).

Recent studies have begun to connect these perturbations directly to action-based potential inference. A. Arora et al. (2022) showed that massive mergers can displace or disrupt stream structures in action space and that time-independent potential representations can lose orbital information on relatively short timescales. Using a time-evolving basis-function-expansion model of the Milky Way–LMC system, R. A. N. Brooks et al. (2024) found that radial-action clustering can retain useful local mass information for shorter-period streams without close LMC encounters, whereas longer-period and strongly perturbed streams may fail to recover the spherically averaged mass profile. These results motivate a distinction between canonical actions calculated in an integrable fitted model and radial-cycle diagnostics constructed from a more realistic but potentially non-integrable potential field.

In this work, we study these limitations using simulated globular-cluster streams evolved in live Milky Way-like halos undergoing mergers with different mass ratios and orbital configurations. For each selected cohort, we maximize the one-dimensional KLD of the canonical radial actions calculated in a trial spherical NFW potential. At the distance-matched snapshot of the fiducial 1:10 merger, the selected cohorts provide a useful recovery of the inner enclosed-mass profile. Repeating the procedure across snapshots and merger strengths then shows when coherent tracers disappear and when surviving cohorts cease to yield stable mass estimates.

We separate these outcomes into three logically distinct levels. First, a perturbed snapshot may contain too few observable coherent cohorts to constrain the potential. Second, even when such cohorts survive, KLD maximization may select a spherical potential different from the cohort-wise target potential. We diagnose this fitted-to-targeted failure through artificial over-clustering and through degeneracies of the KLD score and median action across the trial-parameter plane. Third, even exact recovery of the targeted spherical potential need not reproduce the global spherically averaged mass profile. An exact signed mass-bias decomposition separates the second and third levels.

Our parameter-plane analysis also distinguishes potential-parameter identifiability from mass identifiability. Different NFW parameters may occupy an extended near-maximum KLD ridge while predicting similar enclosed masses over the cohort's radial range. Conversely, a KLD-preferred displacement that crosses enclosed-mass contours produces a mass bias. This directional dependence motivates a practical post-fit screen: map the near-maximum KLD region into enclosed-mass space and favor cohorts for which it corresponds to a narrow mass range. Stream geometry may offer another way to compare KLD-preferred models.

Since several potential models can produce similarly large KLD scores, we also ask whether cohort geometry can help distinguish between them. Motivated by the curvature constraints of J. Nibauer et al. (2023), we compare the measured bend with that implied by a candidate force, restricting the test to cohorts whose mean velocities follow the local track and whose local velocity scatter is small. Under an additional local-orbit approximation, the normal force determines how rapidly a star's direction of motion turns along the track. We therefore test whether agreement with the measured turn improves parameter recovery among models with nearly maximal KLD.

Finally, we use simulation-only information to compare the fitted canonical radial actions with radial-cycle integrations in the frozen full BFE potential and to study the dependence on perturber orbital configuration. These diagnostics do not enter the observable fitting method; they test how the fitted spherical orbital mapping differs from the full simulated field.

Section 2 develops the action and target-potential framework, and Section 3 presents the simulations, cohort selection, and baseline results. Building on these results, Section 4 examines the mass bias and KLD fitting failures before testing the additional bending constraint. Section 5 then presents diagnostics of the full simulated potential, while Section 6 discusses the limitations and future extensions.

2. THEORETICAL FRAMEWORK

2.1. *Canonical Actions and Potential-dependent Action Clustering*

For a Hamiltonian system with n degrees of freedom,the canonical action variables are well defined for regular motion confined to invariant tori. In particular, Liouville integrability requires n functionally independent integrals of motion F_i that are mutually in involution,

$$\{F_i, F_j\} = 0 \qquad \text{for all } i, j. \tag{1}$$

When the corresponding common level set is regular, compact, and connected, it forms an invariant n-torus. The canonical actions are then defined by integrals over the n independent fundamental cycles γ_i of this torus:

$$J_i = \frac{1}{2\pi} \oint_{\gamma_i} \boldsymbol{p} \cdot d\boldsymbol{q}. \tag{2}$$

In separable canonical coordinates, this expression reduces to

$$J_i = \frac{1}{2\pi} \oint_{\gamma_i} p_i \, dq_i. \tag{3}$$

Here, q_i and p_i are canonically conjugate variables. The actions label the invariant torus and remain constant under the integrable Hamiltonian, while the conjugate angles evolve along it.

A static spherically symmetric potential is integrable. Away from degenerate orbits, the energy H, total angular momentum squared L^2, and azimuthal angular momentum L_z provide three functionally independent integrals in involution. Consequently, three canonical actions (J_r, J_θ, J_ϕ) can be defined. Throughout the following expressions, momenta and energies are taken per unit stellar mass. The energy is

$$\begin{aligned} E &= \frac{1}{2}\left(p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2 \sin^2\theta}\right) + \Phi(r) \\ &= \frac{1}{2}\left(p_r^2 + \frac{L^2}{r^2}\right) + \Phi(r), \end{aligned} \tag{4}$$

where

$$L^2 = p_\theta^2 + \frac{p_\phi^2}{\sin^2\theta}. \tag{5}$$

Since the azimuthal angle ϕ is cyclic in a spherical potential, its conjugate momentum is conserved:

$$p_\phi = L_z. \tag{6}$$

So that the azimuthal action is:

$$J_\phi = \frac{1}{2\pi} \oint_l p_\phi \cdot d\phi = p_\phi = L_z \tag{7}$$

Using the expression of angular momentum shown in Equ.4, the polar action can be expressed as:

$$J_\theta = \frac{1}{2\pi} \oint_l p_\theta \cdot d\theta = \frac{1}{\pi} \int_{\theta_{min}}^{\theta_{max}} \sqrt{L^2 - \frac{L_z^2}{\sin^2\theta}} \cdot d\theta \tag{8}$$

Using $\sin(\theta_{min}) = \sin(\theta_{max}) = \frac{|L_z|}{L}$, the integral gives:

$$J_\theta = L - |L_z| \tag{9}$$

Finally, using Equ.4, the radial action can also be acquired:

$$J_r = \frac{1}{\pi} \int_{r_{min}}^{r_{max}} \sqrt{2[E - V(r)] - \frac{L^2}{r^2}} \cdot dr \tag{10}$$

where $r_{\min}$ and $r_{\max}$ are the radial turning points in the adopted spherical potential. For the fitted static spherical halo models, we calculate these canonical actions using `Agama`. The code determines the allowed radial interval and evaluates the corresponding canonical actions.

Stars stripped from a dynamically cold progenitor begin on nearby orbits and therefore occupy a relatively compact region in action space when evaluated in an appropriate integrable potential. This compactness contains information about the potential. Stars from the same stream generally sample different orbital phases. *An appropriate potential maps these different phase-space points onto nearby invariant tori,* whereas an incorrect trial potential generally assigns phase-dependent

action errors, broadening or distorting the inferred action cloud. This is the physical basis of action-clustering potential inference (R. E. Sanderson et al. 2015; S. Reino et al. 2021, 2022).

This statement is not an identity: a wrong potential may translate an action cloud without greatly broadening it, and a short stream segment may provide insufficient phase coverage to distinguish trial models. Potential recovery therefore requires both a dynamically cold tracer and sufficient diversity in orbital phase, radius, energy, or angular momentum. Combining multiple streams on different orbits can reduce individual-stream degeneracies and biases (R. E. Sanderson et al. 2015; S. Reino et al. 2021).

2.2. *1D Kullback-Leibler Divergence*

We quantify radial-action clustering using the Kullback–Leibler divergence (KLD). For a radial-action probability density $p(J_r)$ and a reference density $q(J_r)$,

$$D_{\mathrm{KL}}[p\|q] = \int p(J_r)\ln\left[\frac{p(J_r)}{q(J_r)}\right] dJ_r. \tag{11}$$

where $p(J_r)$ is the probability density of the radial action distribution at the points J_r, and $q(J_r)$ is the background distribution.

Also, following the work by R. E. Sanderson et al. (2015), I use the Modified Breiman Density Estimator to calculate the density profile of the action points. According to B. J. Ferdosi et al. (2011), a fine grid will be built across the action space, and the density value at the center of each fine grid $J_{r,k}$ will be evaluated using the following formula:

$$p(J_{r,k}) = \frac{1}{N}\Sigma_{i=1}^{N}(\sigma\lambda_i)^{-d}K_E(\frac{J_{r,k}-J_{r,i}}{\sigma_r\lambda_i}) \tag{12}$$

Here, i is the star index, N stands for the total number of stars, and d stands for the dimensionality of the action space. This estimator uses the Epanechnikov Kernel K_E instead of a Gaussian Kernel:

$$K_E(t) = \begin{cases} \frac{d+2}{2V_d}(1-t^2), & \text{if } t^2<1, \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

, where V_d is the volume of the unit sphere in d-dimensional space. We can easily see the the kernel contribution decreases with normalized distance from the sample point to the grid center, which is as expected. Following the analysis in B. J. Ferdosi et al. (2011), the window sizes σ_j in each direction in the action space is derived from coordinate percentiles in each dimension in the action space:

$$\sigma_r = \min\{\frac{\mathrm{P}_{80}(J_{r,i})-\mathrm{P}_{20}(J_{r,i})}{\ln\mathrm{N}}\} \tag{14}$$

, where $\mathrm{P}_n(J_{r,i})$ is the nth percentile of the radial action values. In addition, define a pilot density $\mathrm{p}_{\mathrm{pilot}}(J_{r,i})$ to describe the degree of clustering for the data point by using Equ. 12, with $\lambda_i = 1$. And the adaptive bandwidth λ_i of this estimator is designed to be inversely proportional to the $\mathrm{p}_{\mathrm{pilot}}(J_{r,i})$:

$$\lambda_i = (\frac{\mathrm{p}_{\mathrm{pilot}}(J_{r,i})}{g})^{-\alpha} \tag{15}$$

, where g is the geometric mean of the pilot density:

$$g = \exp[\ \frac{1}{N}\ \Sigma_{i=1}^{N}\ln\ \mathrm{p}_{\mathrm{pilot}}(J_{r,i})\ ] \tag{16}$$

and $\alpha = \frac{1}{d}$ is the adaptive-bandwidth exponent. This adaptive bandwidth shrinks kernels in dense regions and widens them in sparse regions, thus increasing the density value where the action points cluster tightly.

Also, for those grids with overly small $p(J_r)$ values, a floor value (like 10^{-9}) needs to be assigned to prevent error. Luckily, the function "`modified_breiman_density`" in the "`MBEtree`" package can realize the above calculation for $p(J_r)$.

Following the action-clustering framework of R. E. Sanderson et al. (2015) and its applications by S. Reino et al. (2021, 2022), trial gravitational potentials are compared through the degree of clustering of the actions calculated from the same observed phase-space data. In this work, the reference distribution $q(J_r)$ is uniform over one fixed and sufficiently wide action interval $[J_{r,\min}, J_{r,\max}]$ that is shared by every trial potential,

$$q(J_r) = \frac{1}{J_{r,\max}-J_{r,\min}}. \tag{17}$$

The interval $[J_{r,\min}, J_{r,\max}]$ is kept the same for all trial potentials. In practice, we choose this universal range to be sufficiently wide to contain the radial actions produced over the full set of potentials considered, including deliberately extreme choices of the potential parameters. This ensures that changes in the KLD are not driven by changing the support of the background distribution from one potential to another.

Thus, the KLD measures how strongly the radial-action distribution departs from a uniform distribution over the fixed reference range. Since the reference interval is held fixed for all trial potentials, maximizing D_{KL}^{I} is equivalent to finding the potential in which the stream is most concentrated in J_r, or equivalently, has the lowest radial-action entropy. A mismatched potential generally broadens the inferred J_r distribution of a stream, thereby reducing the KLD. The KLD test used here is based on the expectation that the most accurate potential produces the sharpest radial-action clustering relative to the same uniform background. The success of this method has been demonstrated in static test potentials and in applications combining multiple stellar

streams (R. E. Sanderson et al. 2015; S. Reino et al. 2021, 2022).

2.3. *Cohort-wise Snapshot target potentials*

A live merger differs fundamentally from the static integrable setting. Its true Hamiltonian,

$$H(\boldsymbol{z},t) = \frac{1}{2}|\boldsymbol{p}|^2 + \Phi(\boldsymbol{x},t), \qquad \boldsymbol{z} \equiv (\boldsymbol{x},\boldsymbol{p}), \tag{18}$$

is time dependent and generally non-integrable. The true trajectories therefore need not lie on invariant three-tori, and the full merger potential does not generally possess a global set of conserved canonical actions.

The fitting problem nevertheless requires an integrable model that can assign actions to the observed phase-space points. We therefore define a theoretical spherical target potential separately for every material cohort C and every instant t, whose force field best represent the acceleration field in the region sampled by the cohort C:

$$\Phi_{\mathrm{tar},C}(r,t) = \Phi_{\mathrm{sph}}[r;\boldsymbol{\lambda}_{\mathrm{tar},C}(t)]\,. \tag{19}$$

Here C denotes a fixed set of stellar members when its history is considered. Different cohorts generally occupy different regions of the force field and therefore possess different time-indexed sequences of targeted parameters.

In the simulations, a precise theoretical definition is obtained by minimizing the force mismatch over the cohort,

$$\boldsymbol{\lambda}_{\mathrm{tar},C}(t) = \arg\min_{\boldsymbol{\lambda}} \frac{1}{N_C}\sum_{a\in C}\left\|\boldsymbol{g}_{\mathrm{true}}[\boldsymbol{x}_a(t),t] - \boldsymbol{g}_{\mathrm{sph}}[r_a(t);\boldsymbol{\lambda}]\right\|^2, \tag{20}$$

where $\boldsymbol{g}_{\mathrm{true}} = -\nabla\Phi_{\mathrm{true}}$ represents the actual local acceleration, $\boldsymbol{g}_{\mathrm{sph}} = -\nabla\Phi_{\mathrm{sph}}$ is the acceleration under the spherical model parametrized by $\boldsymbol{\lambda}$. Because every candidate is spherical, only its radial force can be adjusted; tangential residual forces remain part of the mismatch.

At a snapshot t_n, freezing any candidate value of $\boldsymbol{\lambda}$ gives a static integrable Hamiltonian,

$$H_{\mathrm{sph}}(\boldsymbol{z};\boldsymbol{\lambda}) = \frac{1}{2}|\boldsymbol{p}|^2 + \Phi_{\mathrm{sph}}(r;\boldsymbol{\lambda}), \tag{21}$$

which assigns a regular model orbit and canonical actions $\boldsymbol{J}(\boldsymbol{z};\boldsymbol{\lambda})$ to each stellar phase-space point. The snapshot-wise target action of member a is consequently

$$J^{(a)}_{\mathrm{tar},C,i}(t) \equiv J_i[\boldsymbol{z}_a(t);\boldsymbol{\lambda}_{\mathrm{tar},C}(t)]\,. \tag{22}$$

It is a canonical action of the instantaneously frozen targeted Hamiltonian, but it is not generally conserved along the star's actual trajectory.

For a given material cohort, Equations 20 and 22 define the snapshot sequence

$$\Phi_{\mathrm{tar},C}(t_1) \to \Phi_{\mathrm{tar},C}(t_2) \to \cdots \to \Phi_{\mathrm{tar},C}(t_n). \tag{23}$$

This sequence is a chain of cohort-wise spherical approximations, not the physical evolution of an integrable true potential. If the best-matching parameter values change smoothly with time, this sequence may be represented locally by a differentiable function $\boldsymbol{\lambda}_{\mathrm{tar},C}(t)$, allowing the continuous-time action-evolution equation derived below. If the best-matching parameters instead jump between distinct solutions, $\dot{\boldsymbol{\lambda}}_{\mathrm{tar},C}(t)$ is not defined at the transition. The exact neighboring-snapshot decomposition remains valid, but the change in the target action map derived below must then be treated as a finite increment.

2.4. *Hamiltonian Evolution of the Assigned target actions*

For brevity, consider one member a of cohort C and suppress the labels a and C temporarily. The assigned action is the composite function

$$J_{\mathrm{tar},i}(t) = J_i[\boldsymbol{z}(t);\boldsymbol{\lambda}_{\mathrm{tar}}(t)]\,. \tag{24}$$

Its total derivative along the actual stellar trajectory is

$$\frac{dJ_{\mathrm{tar},i}}{dt} = \left.\frac{\partial J_i}{\partial \boldsymbol{z}}\right|_{\boldsymbol{\lambda}_{\mathrm{tar}}}\cdot\dot{\boldsymbol{z}} + \dot{\lambda}^{\alpha}_{\mathrm{tar}}\left.\frac{\partial J_i}{\partial \lambda^{\alpha}_{\mathrm{tar}}}\right|_{\boldsymbol{z}}. \tag{25}$$

The first term is the change caused by the star's actual phase-space motion while the instantaneous target potential is held fixed. The second is the change caused by updating that map at a fixed physical phase-space point. Repeated target-potential parameter indices α are summed.

The first term follows directly from Hamilton's equations. Holding $\boldsymbol{\lambda}_{\mathrm{tar}}$ fixed, the first term becomes:

$$\begin{aligned}\left.\frac{dJ_i}{dt}\right|_{\boldsymbol{\lambda}_{\mathrm{tar}}} &= \sum_k\left(\frac{\partial J_i}{\partial x_k}\dot{x}_k + \frac{\partial J_i}{\partial p_k}\dot{p}_k\right)\\ &= \sum_k\left(\frac{\partial J_i}{\partial x_k}\frac{\partial H}{\partial p_k} - \frac{\partial J_i}{\partial p_k}\frac{\partial H}{\partial x_k}\right) = \{J_i,H\},\end{aligned} \tag{26}$$

where $\dot{x}_k = \partial H/\partial p_k$ and $\dot{p}_k = -\partial H/\partial x_k$ are generated by the true Hamiltonian.

At that instant, decompose the true Hamiltonian into the frozen targeted Hamiltonian and a residual,

$$H(\boldsymbol{z},t) = \frac{1}{2}|\boldsymbol{p}|^2 + \Phi_{\mathrm{tar},C}[r;\boldsymbol{\lambda}_{\mathrm{tar},C}(t)] + \delta\Phi_C \tag{27}$$

$$= H_{\mathrm{tar},C}[\boldsymbol{z};\boldsymbol{\lambda}_{\mathrm{tar},C}(t)] + \delta\Phi_C, \tag{28}$$

For canonical action–angle coordinates $(\boldsymbol{\theta},\boldsymbol{J})$, the Poisson bracket of two phase-space functions J_i and B is:

$$\{J_i,B\} = \sum_k\left(\frac{\partial J_i}{\partial\theta_k}\frac{\partial B}{\partial J_k} - \frac{\partial J_i}{\partial J_k}\frac{\partial B}{\partial\theta_k}\right). \tag{29}$$

Also, the canonical action coordinates satisfy

$$\frac{\partial J_i}{\partial\theta_k} = 0, \qquad \frac{\partial J_i}{\partial J_k} = \delta_{ik}, \tag{30}$$

where δ_{ik} is the Kronecker delta. So we have:

$$\begin{aligned}\{J_i, B\} &= -\sum_k \delta_{ik}\frac{\partial B}{\partial \theta_k} \\ &= -\frac{\partial B}{\partial \theta_i}.\end{aligned} \tag{31}$$

Therefore,

$$\begin{aligned}\{J_i, H\} &= \{J_i, H_{\mathrm{tar},C}\} + \{J_i, \delta\Phi_C\} \\ &= -\frac{\partial H_{\mathrm{tar},C}}{\partial \theta_{\mathrm{tar},C,i}} - \frac{\partial \delta\Phi_C}{\partial \theta_{\mathrm{tar},C,i}}.\end{aligned} \tag{32}$$

And since in the action–angle coordinates of the frozen targeted Hamiltonian, $H_{\mathrm{tar},C} = H_{\mathrm{tar},C}(\boldsymbol{J}; \boldsymbol{\lambda}_{\mathrm{tar},C})$ has no angle dependence, the above equation can be reduced to:

$$\{J_i, H\} = -\frac{\partial \delta\Phi_C}{\partial \theta_{\mathrm{tar},C,i}}. \tag{33}$$

Now we define the residual force

$$\delta \boldsymbol{F}_C \equiv \boldsymbol{g}_{\mathrm{true}} - \boldsymbol{g}_{\mathrm{tar},C} = -\nabla \delta\Phi_C, \tag{34}$$

where $\boldsymbol{g}_{\mathrm{tar},C} \equiv -\nabla\Phi_{\mathrm{tar},C}$, the chain rule gives

$$\begin{aligned}-\frac{\partial \delta\Phi_C}{\partial \theta_{\mathrm{tar},C,i}} &= -\nabla\delta\Phi_C \cdot \frac{\partial \boldsymbol{x}}{\partial \theta_{\mathrm{tar},C,i}} \\ &= \delta \boldsymbol{F}_C \cdot \frac{\partial \boldsymbol{x}}{\partial \theta_{\mathrm{tar},C,i}}.\end{aligned} \tag{35}$$

Restoring the cohort and member labels in Equation 25 yields

$$\boxed{\frac{dJ^{(a)}_{\mathrm{tar},C,i}}{dt} = \delta \boldsymbol{F}^{(a)}_C \cdot \frac{\partial \boldsymbol{x}}{\partial \theta_{\mathrm{tar},C,i}} + \dot{\lambda}^{\alpha}_{\mathrm{tar},C}\left.\frac{\partial J^{(a)}_{\mathrm{tar},C,i}}{\partial \lambda^{\alpha}_{\mathrm{tar},C}}\right|_{\boldsymbol{z}_a}}. \tag{36}$$

Equation 36 describes the instantaneous changing rate of the assigned target action. If the target-potential parameters vary differentiably, the finite action change between two snapshots can be obtained by integrating this rate over the intervening time interval. For sufficiently closely spaced snapshots, the corresponding finite-difference rate provides an approximation to the instantaneous rate.

To show how the two contributions in Equation 36 appear in the snapshot-wise description, consider the assigned-action change between neighboring snapshots. Subtracting and adding the action assigned to the new phase-space point by the earlier target potential gives:

$$\begin{aligned}&J_i(\boldsymbol{z}_{a,n+1}; \boldsymbol{\lambda}_{\mathrm{tar},C,n+1}) - J_i(\boldsymbol{z}_{a,n}; \boldsymbol{\lambda}_{\mathrm{tar},C,n}) \\ &= \Big[J_i(\boldsymbol{z}_{a,n+1}; \boldsymbol{\lambda}_{\mathrm{tar},C,n}) - J_i(\boldsymbol{z}_{a,n}; \boldsymbol{\lambda}_{\mathrm{tar},C,n})\Big] \\ &\quad + \Big[J_i(\boldsymbol{z}_{a,n+1}; \boldsymbol{\lambda}_{\mathrm{tar},C,n+1}) - J_i(\boldsymbol{z}_{a,n+1}; \boldsymbol{\lambda}_{\mathrm{tar},C,n})\Big].\end{aligned} \tag{37}$$

Equation 37 is an exact decomposition of the finite action change. The first bracket evaluates the star's two actual phase-space points using the same, earlier target potential at snapshot n. After division by Δt, it approaches the residual-force contribution in Equation 36:

$$\begin{aligned}&\frac{J_i(\boldsymbol{z}_{a,n+1}; \boldsymbol{\lambda}_{\mathrm{tar},C,n}) - J_i(\boldsymbol{z}_{a,n}; \boldsymbol{\lambda}_{\mathrm{tar},C,n})}{\Delta t} \\ &\xrightarrow{\Delta t\to 0} \delta \boldsymbol{F}^{(a)}_C \cdot \frac{\partial \boldsymbol{x}}{\partial \theta_{\mathrm{tar},C,i}}.\end{aligned} \tag{38}$$

The second bracket evaluates the same ending phase-space point using the target potentials at the two snapshots $n+1$ and n. A first-order Taylor expansion in the target-potential parameters gives

$$\begin{aligned}&\frac{J_i(\boldsymbol{z}_{a,n+1}; \boldsymbol{\lambda}_{\mathrm{tar},C,n+1}) - J_i(\boldsymbol{z}_{a,n+1}; \boldsymbol{\lambda}_{\mathrm{tar},C,n})}{\Delta t} \\ &\xrightarrow{\Delta t\to 0} \dot{\lambda}^{\alpha}_{\mathrm{tar},C}\left.\frac{\partial J^{(a)}_{\mathrm{tar},C,i}}{\partial \lambda^{\alpha}_{\mathrm{tar},C}}\right|_{\boldsymbol{z}_a}.\end{aligned} \tag{39}$$

Thus, the two finite changes in Equation 37 correspond to the same two physical effects represented by Equation 36. For a finite snapshot interval, their ratios to Δt represent average action-changing rates. They approach the instantaneous terms in Equation 36 in the limit $\Delta t \to 0$.

2.5. *From target-action Evolution to Cohort Selection*

The action cloud remains useful for clustering only if the evolution in Equation 36 is sufficiently similar among the members. For two members a and b,

$$\begin{aligned}\frac{d}{dt}\left(J^{(a)}_{\mathrm{tar},C,i} - J^{(b)}_{\mathrm{tar},C,i}\right) &= \Delta_{ab}\left[\delta \boldsymbol{F}_C \cdot \frac{\partial \boldsymbol{x}}{\partial \theta_{\mathrm{tar},C,i}}\right] \\ &\quad + \dot{\lambda}^{\alpha}_{\mathrm{tar},C}\Delta_{ab}\left[\left.\frac{\partial J_{\mathrm{tar},C,i}}{\partial \lambda^{\alpha}_{\mathrm{tar},C}}\right|_{\boldsymbol{z}}\right],\end{aligned} \tag{40}$$

where $\Delta_{ab}X \equiv X^{(a)} - X^{(b)}$. Integrating between an earlier time t_1 and the observed time t_0 gives

$$\Delta_{ab}J_{\mathrm{tar},C,i}(t_0) = \Delta_{ab}J_{\mathrm{tar},C,i}(t_1) + \int_{t_1}^{t_0} \Delta_{ab}\dot{J}_{\mathrm{tar},C,i}(t)\, dt. \tag{41}$$

If the total action change is nearly common across the cohort, the cloud may translate through action space while retaining its internal compactness. If the changes are strongly member dependent, the cloud broadens or distorts even under the correct instantaneous target potential. KLD maximization can then prefer a different trial potential that artificially compresses the disturbed distribution.

The force history and the sequence $\Phi_{\mathrm{tar},C}(t)$ are not directly observable. We therefore use present-day phase-space structure as evidence, rather than proof, of a semi-rigid past evolution. A cohort that is smooth in velocity,

occupies one unfolded phase-space branch, spans a restricted region, and lies away from a radial turning point is more likely to have experienced similar residual-force projections and similar action-map changes among its members. Passing the selection at the observed time t_0 is consequently a strong indicator that the instantaneous target-action cloud may remain compact enough to support the action-clustering inference.

The selected cohort population may differ from one observed snapshot to another without invalidating this construction. Every cohort selected at a given snapshot t_0 has its own material history and its own hypothetical sequence of cohort-wise target potentials. The time-dependent population plots later in the paper are repeated cross-sectional applications of the same set of criteria; they do not assert that one fixed cohort survives the selection at every snapshot.

3. SIMULATIONS, COHORT SELECTION, AND BASELINE PERFORMANCE

3.1. *Merger Simulations and BFE Representation*

We use the stellar-stream simulations from *Dancing Streams in Merging Halos* (Weerasooriya et al. 2025), a controlled suite designed to study how globular-cluster streams respond to a Milky Way–LMC-like merger. The simulations first generate 1024 mock globular-cluster streams in a static Milky Way-like Hernquist halo using the Fardal et al. (2015) mock-stream distribution function as implemented in `Gala`. Each stream contains 1000 stellar particles. The grid spans progenitor masses of 10^4 and $10^6\,M_\odot$, eight apocentric radii from 30 to 100 kpc, eight circularities from $\eta = 0.5$ to 1.0, and eight stream ages from 0.5 to 6 Gyr.

The same stream population is then inserted into live N-body halos and evolved with `Gadget4`. The host halo has mass $M_{\rm halo} = 1.57 \times 10^{12}\,M_\odot$ and Hernquist scale radius $a = 40.85\,$kpc. In addition to the published isolated and 1:5 simulations, we use related data products spanning perturber-to-host mass ratios 1:2, 1:5, 1:10, and 1:100, together with standard, more circular, and more radial 1:5 orbital configurations. The simulations contain 241 snapshots over 6 Gyr; the principal KLD analysis uses snapshots 000–100, corresponding to approximately 0–2.56 Gyr.

The time-dependent halo field is represented by a basis-function expansion (BFE),

$$\rho(\boldsymbol{x}, t) \simeq \sum_{n,\ell,m} a_{n\ell m}(t)\, u_{n\ell}(r) Y_{\ell m}(\theta, \phi), \qquad (42)$$

with the corresponding potential obtained from Poisson's equation. The adopted coefficients were supplied by David Amir Gonzalez and Robyn Sanderson and accurately represent the later-stage potential over the BFE-valid intervals used below. The BFE provides the simulated full force field required to define the cohort-wise target potentials and the simulation-only diagnostics; it is not used as an observable input to the KLD fit.

3.2. *Observable Cohort-selection Criteria*

Neither the residual-force history nor the target-potential chain is available in an observation. We therefore implement the theoretical motivation of Section 2.5 through present-day phase-space criteria applied before the KLD fit.

First, we require a smooth velocity profile along the stream. For an along-stream coordinate s, we fit

$$v_{\rm fit}(s) = As^2 + Bs + C \qquad (43)$$

and retain only cohorts satisfying

$$\Delta_v \equiv \frac{\sqrt{\langle [v_\parallel(s) - v_{\rm fit}(s)]^2 \rangle}}{\mathrm{std}[v_\parallel(s)]} < 0.1. \qquad (44)$$

A smooth profile is consistent with a small member-dependent accumulated velocity response, although it does not uniquely determine the past force history.

Second, we require an unfolded, single-valued radial phase-space branch. Operationally, $v_{\rm fit}(s)$ must remain strictly monotonic over the selected interval. This rejects folded structures in which nearby positions correspond to different orbital branches and therefore potentially different targeted-orbit tangents and action sensitivities.

Third, the cohort must have a limited spatial extent, so that its members sample a restricted region of the force field and can be represented by one cohort-common target potential. An extremely narrow radial segment, however, has little leverage to distinguish trial force profiles. We therefore additionally require

$$\Delta r_{\rm cohort} \equiv \max_a(r_a) - \min_a(r_a) \geq 2\ \text{kpc}. \qquad (45)$$

The upper extent restriction is a coherence condition, whereas the lower radial-span restriction is an identifiability condition. All present-day candidate cohorts considered here already satisfy the latter.

Fourth, we exclude cohorts close to radial turning points by imposing a threshold on $|v_r|$. Near pericentre or apocentre, the mapping between local phase-space coordinates and radial phase becomes poorly conditioned, and incoming and outgoing branches can have substantially different responses despite occupying similar radii.

Finally, every retained cohort must contain at least 100 stars. The four physical criteria are thus smooth along-stream velocity, an unfolded phase-space branch, limited spatial extent with sufficient radial span, and separation from radial turning points. Figure 1 illustrates that this procedure is deliberately restrictive: it retains short, nearly single-valued segments rather than merely removing isolated outliers. Passing these criteria favors, but does not guarantee, a compact target-action cloud and reliable potential recovery.

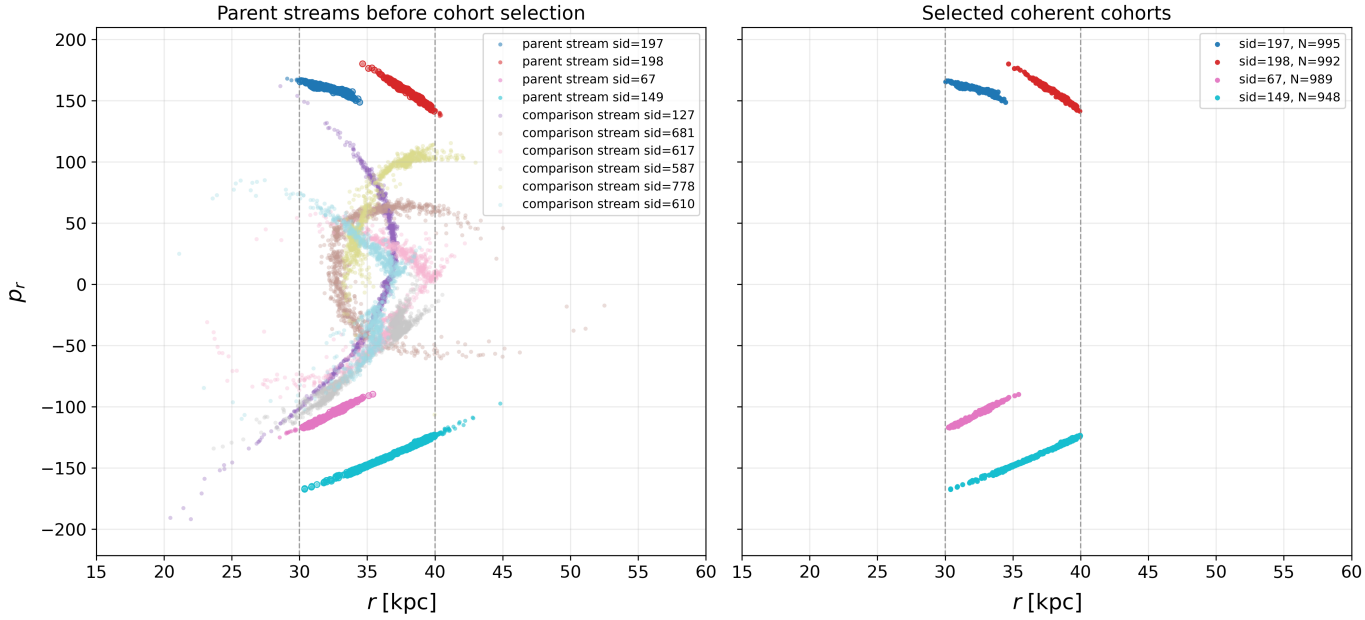


Figure 1. Illustration of the cohort-selection procedure in radial phase space for the 1:10 merger at snapshot 030 in the $30 \leq r/\mathrm{kpc} \leq 40$ radial range. The vertical dashed lines mark the boundaries of the selected radial interval. **Left:** four parent streams together with six comparison streams before cohort selection, including extended, folded, and kinematically irregular phase-space structures. **Right:** the locally coherent segments retained from the four parent streams after requiring a smooth along-stream velocity profile, an unfolded monotonic radial phase-space branch, limited spatial extent, and separation from radial turning points. The listed values of N give the stellar membership of each retained cohort. These criteria favor segments with smaller member-to-member differential forcing and targeted-coordinate remapping, but do not by themselves guarantee an unbiased potential fit.

3.3. *Implementation and Distance-matched Present-day Recovery*

We first implement the selection in the fiducial 1:10 merger. Because the LMC is currently at a Galactocentric distance of approximately 49.5 kpc (X. Wu et al. 2008) and is likely on its first infall or first passage (G. Besla et al. 2007; N. Kallivayalil et al. 2013), we identify snapshot 55 (t = 1.406 Gyr) as the distance-matched present-day analogue. At this snapshot the simulated host–perturber separation first crosses the observed value.

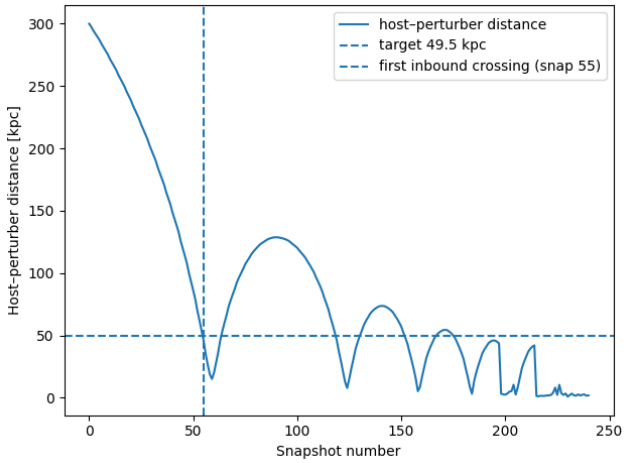


Figure 2. Host–perturber separation as a function of snapshot number for the fiducial Milky Way–LMC-like merger. The horizontal dashed line marks the target separation of 49.5 kpc, and the vertical dashed line identifies its first inbound crossing at snapshot 055 (t = 1.406 Gyr). We adopt this snapshot as the distance-matched present-day analogue used for the single-snapshot cohort-selection and enclosed–mass analysis.

For every selected cohort, we maximize the radial-action KLD over a spherical NFW family and obtain an independent enclosed-mass profile. The initial halo is spherical, but the live merger field later becomes non-spherical; the fitted NFW potential is therefore an action-computable approximation, not a claim that the true halo remains spherical. We retain cohorts containing at least 100 stars in successive 10-kpc-wide radial ranges. No cohort passes the adopted criteria inside 10 kpc, where the simulated perturbations are especially strong.

The spherical enclosed mass implied by an arbitrary gravitational potential can be obtained from the radial gravitational acceleration on a spherical surface. The gravitational acceleration is

$$\boldsymbol{g}(\boldsymbol{x},t) = -\boldsymbol{\nabla}\Phi(\boldsymbol{x},t), \tag{46}$$

while Poisson's equation gives

$$\nabla^2\Phi(\boldsymbol{x},t) = 4\pi G\rho(\boldsymbol{x},t). \tag{47}$$

Combining Equations 46 and 47, we obtain

$$\boldsymbol{\nabla}\cdot\boldsymbol{g}(\boldsymbol{x},t) = -4\pi G\rho(\boldsymbol{x},t). \tag{48}$$

We integrate Equation 48 over the volume V_r enclosed by a sphere of radius r:

$$\begin{aligned}\int_{V_r}\boldsymbol{\nabla}\cdot\boldsymbol{g}\,d^3\boldsymbol{x} &= -4\pi G\int_{V_r}\rho(\boldsymbol{x},t)\,d^3\boldsymbol{x}\\ &= -4\pi G M_{\rm sph}(<r,t).\end{aligned} \tag{49}$$

Applying the divergence theorem gives

$$\oint_{S_r}\boldsymbol{g}\cdot d\boldsymbol{S} = -4\pi G M_{\rm sph}(<r,t), \tag{50}$$

where S_r is the spherical surface bounding V_r.

On a sphere of radius r, the outward surface element is

$$d\boldsymbol{S} = r^2\widehat{\boldsymbol{r}}\,d\Omega, \tag{51}$$

and the radial component of the gravitational acceleration is

$$g_r(r,\Omega,t) \equiv \boldsymbol{g}(r,\Omega,t)\cdot\widehat{\boldsymbol{r}}. \tag{52}$$

Also, we can define the angular average of the radial acceleration at fixed radius as

$$\langle g_r(r,\Omega,t)\rangle_\Omega \equiv \frac{1}{4\pi}\int_{4\pi} g_r(r,\Omega,t)\,d\Omega. \tag{53}$$

Substituting Equations 51, 52 and 53 into Equation 50 yields the spherically averaged enclosed mass:

$$M_{\rm sph}(<r,t) = -\frac{r^2}{G}\,\langle g_r(r,\Omega,t)\rangle_\Omega\,. \tag{54}$$

The above equation does not require the gravitational potential to be spherical. It follows from the total gravitational flux through a spherical surface and therefore returns the mass enclosed by that surface implied by the potential model.

The angular average in Equation 54 is evaluated numerically using $N_\Omega = 4096$ nearly uniformly distributed Fibonacci-sphere directions. Thus,

$$\langle g_r(r,\Omega,t)\rangle_\Omega \simeq \frac{1}{N_\Omega}\sum_{a=1}^{N_\Omega} g_r(r,\Omega_a,t), \tag{55}$$

and the enclosed mass is estimated as

$$M_{\rm sph}(<r,t) \simeq -\frac{r^2}{GN_\Omega}\sum_{a=1}^{N_\Omega} g_r(r,\Omega_a,t). \tag{56}$$

The Fibonacci construction provides nearly equal-area sampling over the sphere and is sufficiently dense relative to the angular resolution of the NFW spherical model.

We combine the independent cohort estimates through a member-number-weighted enclosed-mass profile. In the left panel of Figure 3, we plot the enclosed-mass profile from each cohort only over the radial interval occupied by that cohort. We divide the full radial range into 0.1-kpc-wide bins and, at each bin r_k, average the fitted enclosed masses of the contributing cohorts:

$$M_{\rm fit}(<r_k,t) = \frac{\sum_{i\in\mathcal{I}_k} N_i M_{{\rm sph},i}(<r_k,t)}{\sum_{i\in\mathcal{I}_k} N_i}, \tag{57}$$

where $\mathcal{I}_k$ denotes the selected cohorts whose fitted profiles are evaluated at r_k, and N_i is the number of selected stars in cohort i. The right panel of Figure 3 shows the weight-averaged enclosed mass value at each radial bin from the galactic center outward.

The resulting profile shows that, despite the spherical model simplification, the selected streams recover the enclosed mass reasonably well over $10 \lesssim r/\mathrm{kpc} \lesssim 30$. The discrepancy generally increases farther outward. At snapshot 55, the perturber lies about 49.5 kpc from the host center, and coherent outer-halo segments preferentially survive in less disturbed regions farther from it. A locally constrained spherical model in those regions need not represent the global spherical average and can consequently underestimate the total enclosed mass. At this merger stage, the selected-cohort KLD method is therefore most informative over the inner radial range in which the individual cohort constraints overlap.

3.4. *Time-dependent Inner-halo Performance*

The distance-matched snapshot demonstrates that the selected cohorts can recover a useful enclosed-mass profile at a particular stage of the merger. However, this single-snapshot result does not establish how long the method remains reliable as the halo and stream population evolve. We therefore repeat the cohort selection and spherical-NFW fitting at every snapshot from 0 to 100, corresponding to $0 \le t \lesssim 2.56$ Gyr.

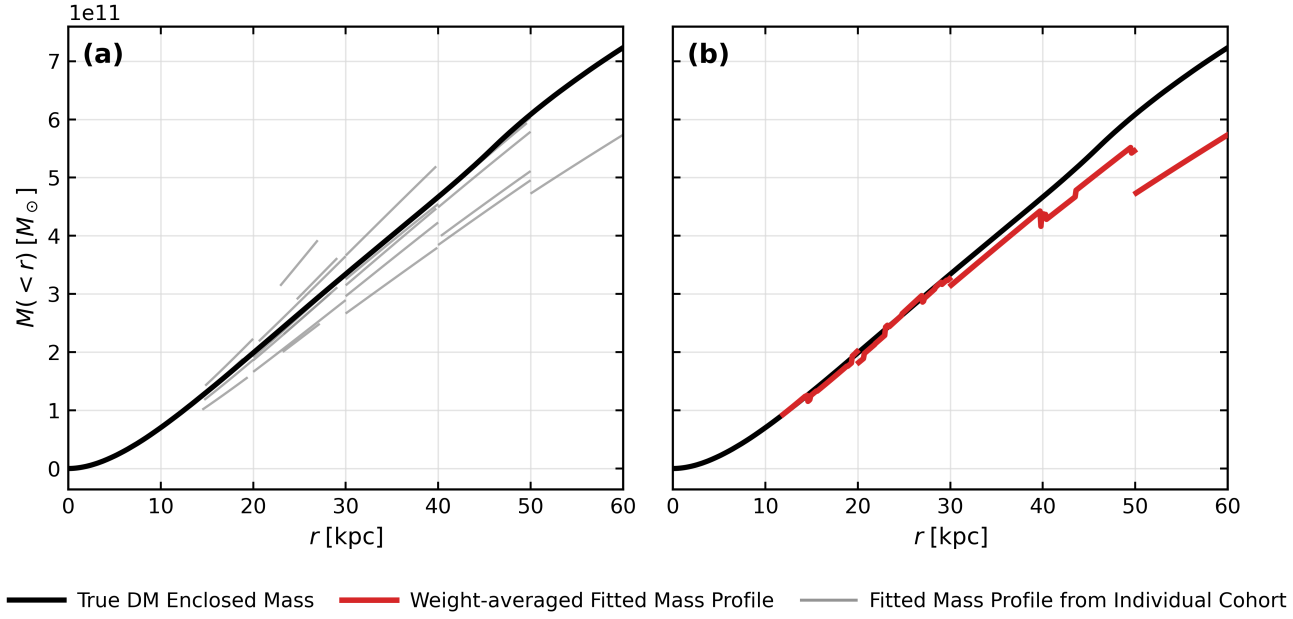


Figure 3. Enclosed-mass recovery from the selected coherent cohorts at snapshot 055, corresponding to the distance-matched present-day configuration of the fiducial 1:10 MW–LMC-like merger. **(a)** KLD-maximized spherical NFW enclosed-mass profiles inferred independently from the individual selected cohorts. Each grey curve is shown only over the radial interval sampled by its corresponding cohort, while the black curve shows the true dark-matter enclosed-mass profile measured directly from the snapshot particles relative to the host center. **(b)** Member-number-weighted mean of the independently fitted cohort mass profiles at each radius (red), compared with the same true enclosed-mass profile (black). Only cohorts whose sampled radial interval includes the given radius contribute to the weighted mean. Thus, the red curve is a radial combination of cohort-specific fits, rather than the result of a single joint fit to all selected stars. The recovered profile agrees most closely with the true profile at smaller radii, while the discrepancy generally increases farther outward. This behavior is consistent with the increasing importance of merger-induced departures from a spherical equilibrium model, together with the decreasing availability and radial coverage of useful coherent cohorts at large radii.

We consider cohorts occupying the radial interval $30 \leq r/\mathrm{kpc} \leq 40$, matching Figure 4. For this inner-halo analysis, the true enclosed mass is measured directly from the host and perturber dark-matter particles, rather than from the BFE potential. At each radial grid point r_k within the 10-kpc-wide radial bin, the weighted fitted enclosed mass is

$$M_{\mathrm{fit}}(< r_k, t) = \frac{\sum_{i \in \mathcal{I}_k} N_i M_{\mathrm{sph},i}(< r_k, t)}{\sum_{i \in \mathcal{I}_k} N_i}, \tag{58}$$

where $\mathcal{I}_k$ denotes the selected cohorts whose fitted profiles are evaluated at r_k. We quantify the signed enclosed-mass bias by

$$b_M(t) = \left\langle \frac{M_{\mathrm{fit}}(< r_k, t) - M_{\mathrm{true}}(< r_k, t)}{M_{\mathrm{true}}(< r_k, t)} \right\rangle_k \times 100\%. \tag{59}$$

Figure 4 summarizes the resulting evolution across all four merger ratios. The first panel shows the number of selected cohorts and their mean stellar membership, the second panel shows $\langle C \rangle_w$, and the third panel shows b_M. The distance-matched present-day comparison time is marked by a light-red vertical line.

The number of selected cohorts generally decreases with time in all four merger cases. However, the cohort-count curves do not show a clear monotonic ordering with merger strength. In particular, the stronger mergers do not consistently lose their selected cohorts earlier than the weaker mergers over the full analyzed interval. The compactness generally increases with time, although it does not exhibit a single threshold that separates successful and unsuccessful fits.

The 1:2 merger loses most of its usable cohorts earliest and develops large, rapidly varying mass biases. The 1:5 and 1:10 cases retain selected material for longer, but subsequently exhibit substantial mass fluctuations. The 1:100 merger remains comparatively stable over most of the analyzed interval. Thus, stronger mergers generally shorten the interval over which the fitted mass remains precise.

Occasional late-time snapshots may nevertheless return a small value of b_M. Such isolated agreement does not by itself demonstrate restored reliability. Because the cohorts are selected independently at each snapshot, the late-time sample can be dominated by an atypical subset that happens to remain especially regular. We

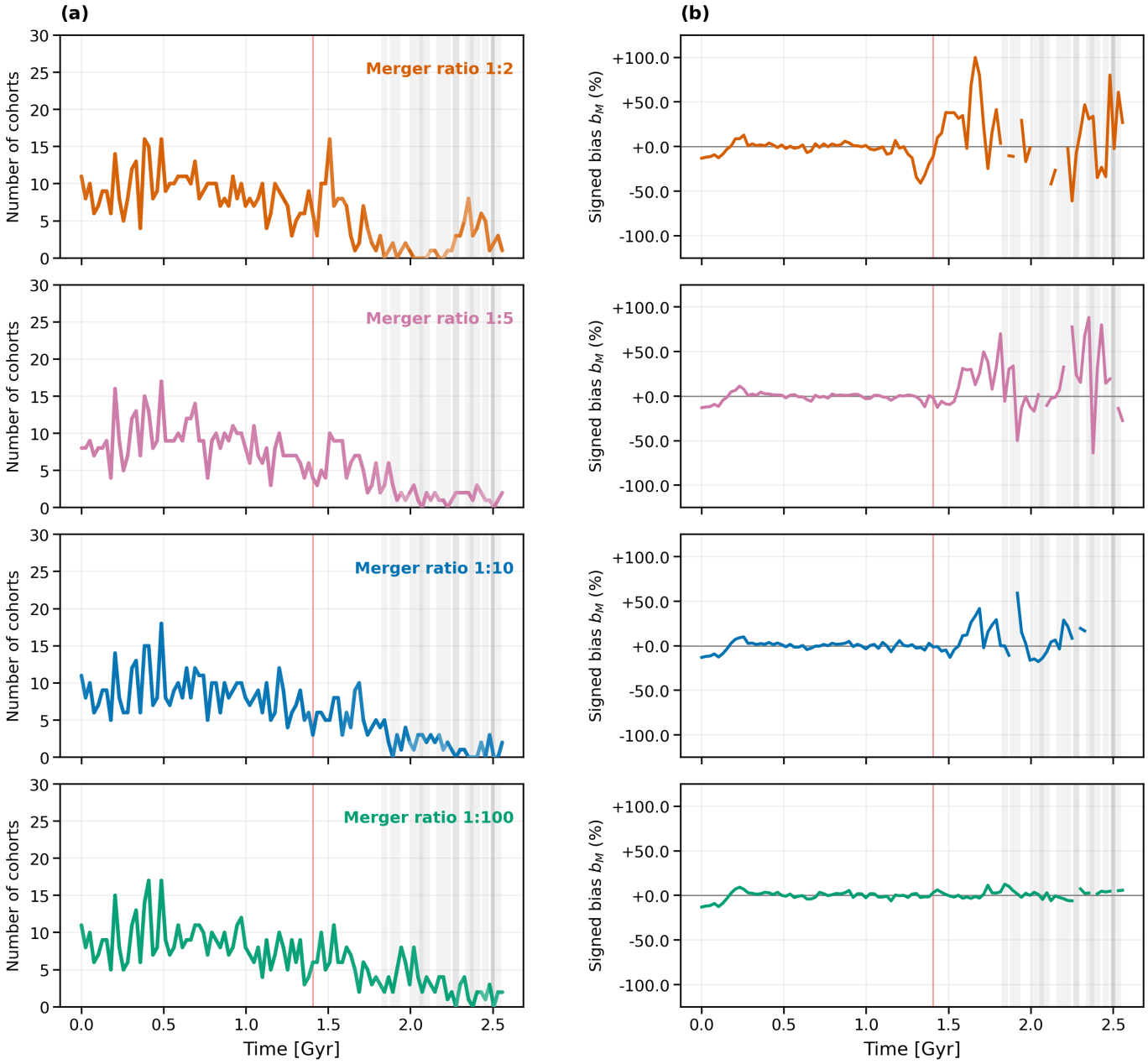


Figure 4. Time-dependent performance of the selected-cohort KLD fitting for merger mass ratios 1:2, 1:5, 1:10, and 1:100 in the $30 \leq r/\mathrm{kpc} \leq 40$ radial range. The light-red vertical line marks the distance-matched present-day comparison time at snapshot 055 ($t = 1.406\,\mathrm{Gyr}$), defined using the fiducial 1:10 merger. **(a)** Number of selected coherent cohorts. Curve color identifies the merger ratio, while the local color saturation encodes the average stellar membership of the selected cohorts; more saturated segments correspond to larger average membership. **(b)** Signed enclosed-mass bias, b_M, of the cohort-member-number-weighted fitted mass profile relative to the true dark-matter enclosed-mass profile measured directly from the snapshot particles and averaged over the radial grid points in the selected range. The background is white when all four merger cases retain at least one selected cohort and becomes progressively darker grey as fewer cases retain usable cohorts. Stronger mergers generally produce larger and more rapidly varying mass biases, whereas the 1:100 merger remains comparatively stable over most of the analyzed interval. The phase-space compactness evolves with time but does not by itself define a universal threshold for successful potential recovery.

therefore interpret the time limit of the method from sustained behavior over multiple snapshots, rather than from individual near-zero mass residuals.

4. POST-SELECTION RELIABILITY AND FAILURE DIAGNOSTICS

The selected-cohort results reveal two distinct ways in which a mass estimate can fail after observable coherence cuts have been applied. The KLD optimizer may select a spherical potential different from the cohort-wise target potential, and the targeted spherical description may itself differ from the global spherically averaged mass profile. We first separate these two levels algebraically. We then examine two non-exclusive reasons for fitted-to-targeted failure: artificial over-clustering of a broadened target-action cloud and degeneracy across the NFW parameter plane. Neither mechanism is an additional additive mass-error component; both can operate within the single fitted-to-targeted term.

4.1. *Decomposition of the Total Enclosed-mass Bias*

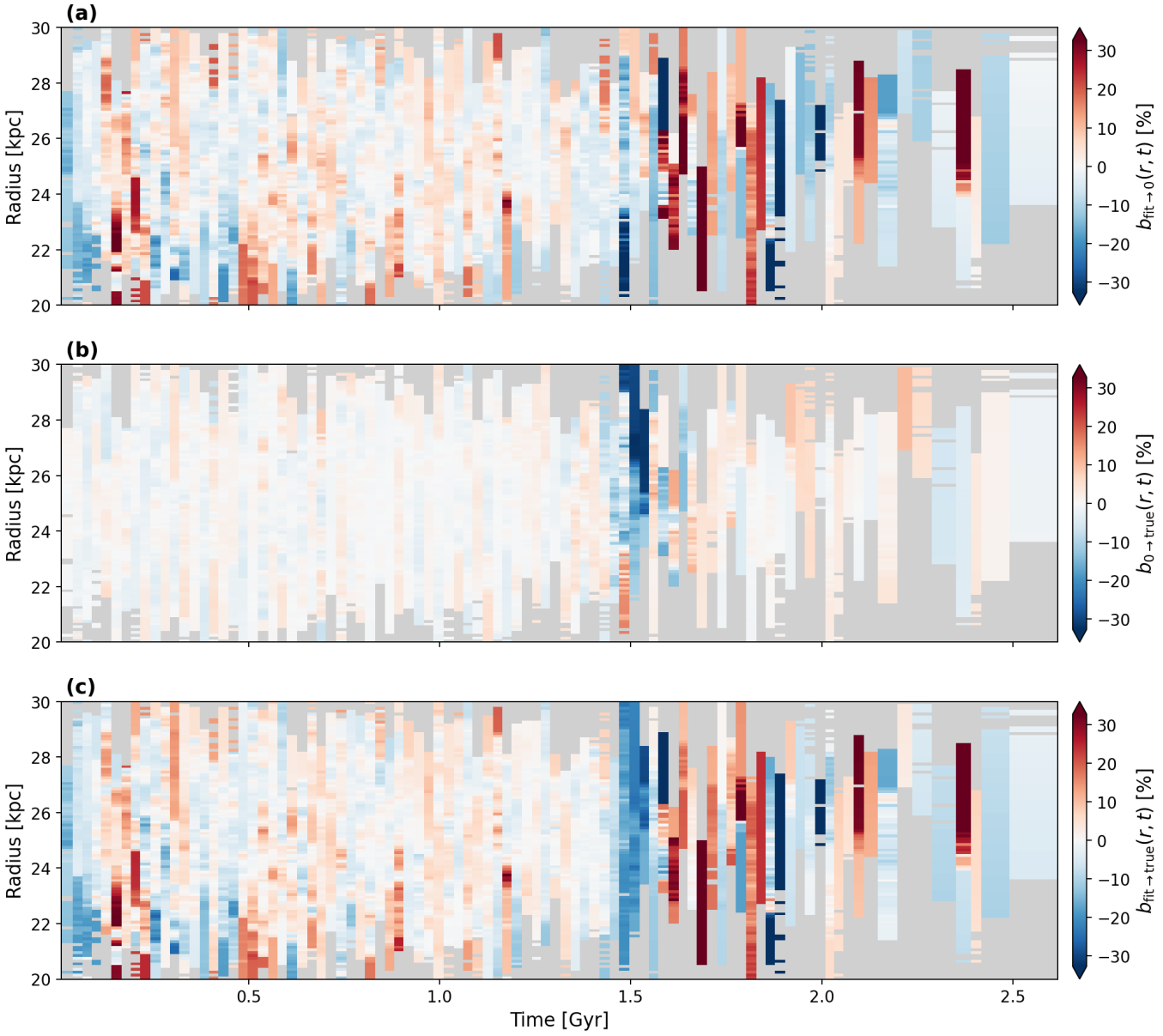


Figure 5. Decomposition of the signed enclosed-mass bias for the 1:10 merger in the $20 \leq r/\mathrm{kpc} \leq 30$ radial range. **(a)** Fitted-to-targeted contribution, $b_{\mathrm{fit}\to\mathrm{tar}}(r,t)$, measuring the difference between the cohort-weighted KLD-best-fit and spherical targeted mass profiles. **(b)** Targeted-to-true contribution, $b_{\mathrm{tar}\to\mathrm{true}}(r,t)$, measuring the difference between the cohort-weighted spherical targeted profile and the globally spherically averaged BFE mass profile. **(c)** Total fitted-to-true bias, $b_{\mathrm{fit}\to\mathrm{true}}(r,t)$. At every valid radius–time pixel, panel **(c)** is

the sum of panels **(a)** and **(b)**. Positive values indicate an overestimate of the enclosed mass, and negative values indicate an underestimate. Gray regions indicate radius–time pixels for which a valid comparison is unavailable. Direct numerical evaluation verifies the additive relation to floating-point precision: the maximum absolute closure residual is 1.510×10^{-14} percentage points and the RMS residual is 1.215×10^{-15} percentage points.

Before examining the action-space mechanisms that can make the KLD fit unreliable, we separate the total enclosed-mass bias into two contributions. The first measures whether the KLD-best-fit spherical potential recovers the spherical target potential. The second measures whether that target potential reproduces the globally spherically averaged mass profile of the full BFE field.

Let $M_{\mathrm{fit},w}(<r,t)$ and $M_{\mathrm{tar},w}(<r,t)$ denote, respectively, the cohort-weighted enclosed masses of the KLD-best-fit and spherical target potentials. Let $M_{\mathrm{true}}(<r,t)$ denote the globally spherically averaged enclosed mass measured from the full BFE model. We define the fitted-to-targeted component as

$$b_{\mathrm{fit}\to\mathrm{tar}}(r,t) = 100\,\frac{M_{\mathrm{fit},w}(<r,t) - M_{\mathrm{tar},w}(<r,t)}{M_{\mathrm{true}}(<r,t)}. \quad (60)$$

This term isolates the mass difference introduced when the KLD fit selects a spherical potential different from the target potential.

The targeted-to-true component is

$$b_{\mathrm{tar}\to\mathrm{true}}(r,t) = 100\,\frac{M_{\mathrm{tar},w}(<r,t) - M_{\mathrm{true}}(<r,t)}{M_{\mathrm{true}}(<r,t)}. \quad (61)$$

This term remains even if the KLD fit recovers the target potential exactly. It measures the difference between the spherical targeted description sampled by the selected cohorts and the globally averaged mass profile of the full BFE field.

The total fitted-to-true mass bias is

$$b_{\mathrm{fit}\to\mathrm{true}}(r,t) = 100\,\frac{M_{\mathrm{fit},w}(<r,t) - M_{\mathrm{true}}(<r,t)}{M_{\mathrm{true}}(<r,t)}. \quad (62)$$

Because all three quantities use the same $M_{\mathrm{true}}(<r,t)$ denominator, they obey the exact additive decomposition

$$b_{\mathrm{fit}\to\mathrm{true}} = b_{\mathrm{fit}\to\mathrm{tar}} + b_{\mathrm{tar}\to\mathrm{true}}. \quad (63)$$

No multiplicative cross-term is required under these definitions.

Figure 5 shows the three terms for the 1:10 merger in the $20 \le r/\mathrm{kpc} \le 30$ radial range. The fitted-to-targeted component in panel **(a)** contains substantial positive and negative structure across radius and time. The largest and most spatially extended deviations become especially prominent after approximately 1.5 Gyr, although localized departures are also present at earlier times. These deviations are the part of the total mass error that can be connected directly to failure of the KLD fit to recover the spherical target potential.

The targeted-to-true component in Figure 5**(b)** is generally smaller and more spatially restricted than the fitted-to-targeted component, but it is not negligible. A particularly coherent negative feature appears around $t \simeq 1.5$ Gyr, while additional localized positive and negative differences occur at later times. These structures show that the spherical target potential does not always reproduce the global BFE enclosed mass even when evaluated over the same selected sample.

The total bias in panel **(c)** visually resembles the fitted-to-targeted component over much of the available radius–time domain. This indicates that failure to recover the target potential is frequently the larger contribution to the total mass-error structure in this example. Nevertheless, panel **(b)** modifies the total bias wherever the targeted-to-true difference is appreciable. The two signed components can reinforce one another when they have the same sign or partially cancel when they have opposite signs. Consequently, a small total bias at an individual pixel does not necessarily imply that both component errors are small.

The direct closure test summarized in the caption of Figure 5 confirms that the signed decomposition is exact to floating-point precision. This test validates the implementation; it does not introduce a third physical source of mass bias.

The decomposition also establishes the scope of the following diagnostics. The target-action spread, KLD over-clustering, KLD-score degeneracy, and median-action degeneracy concern the origin of $b_{\mathrm{fit}\to\mathrm{tar}}$: they test why the KLD procedure may select a spherical potential different from the target potential. They do not directly explain $b_{\mathrm{tar}\to\mathrm{true}}$, which instead reflects the difference between the spherical targeted description and the globally averaged full BFE mass profile. Moreover, the action-space failure mechanisms are not themselves additive mass-error components and may coexist within the single fitted-to-targeted term.

4.2. *Why KLD Maximization Can Miss the target potential*

The KLD method assumes that the cohort-wise target potential is represented by the trial model that produces the strongest radial-action clustering. The following diagnostics show why this implication can fail even after the observable selection has removed strongly irregular material.

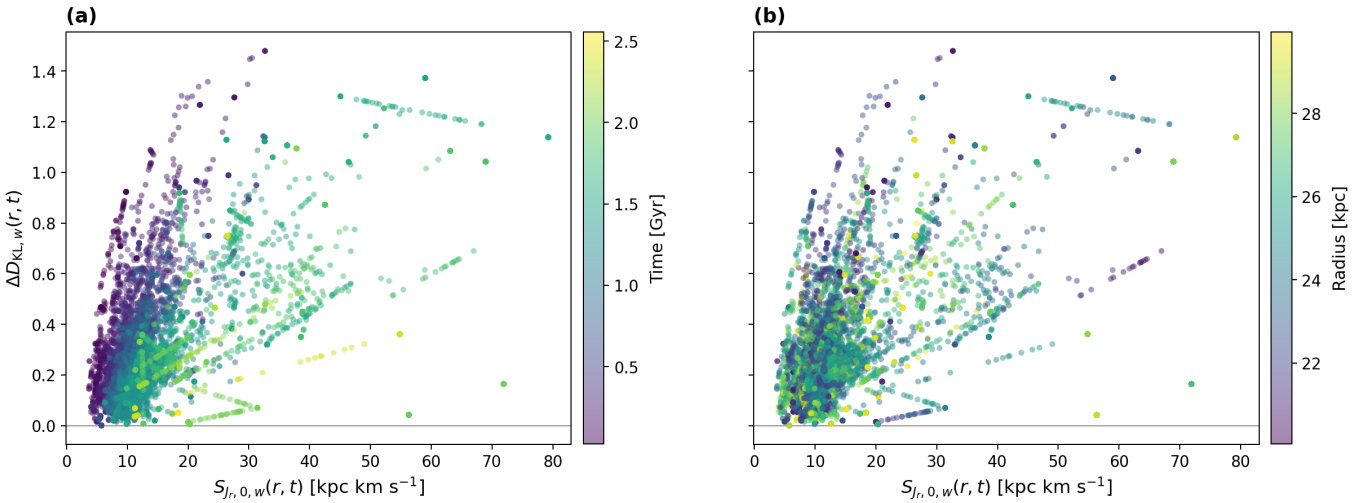


Figure 6. Relation between the weighted targeted radial-action spread, $S_{J_r,\mathrm{tar},w}$, and the weighted KLD excess, $\Delta D_{\mathrm{KL},w}$, for the 1:10 merger in the $20 \leq r/\mathrm{kpc} \leq 30$ radial range. Each point represents one valid radius–snapshot measurement. The points are colored by snapshot time in panel **(a)** and by radius in panel **(b)**. Positive $\Delta D_{\mathrm{KL},w}$ indicates that the KLD fit produces a more strongly clustered action cloud than the target potential. The distribution broadens toward larger KLD excess as the target action cloud becomes more extended, although the relation contains substantial intrinsic scatter. The time coloring in panel (a) shows that earlier snapshots preferentially populate the upper KLD-excess envelope, suggesting that their broad action clouds remain more readily compressible by a mismatched trial potential.

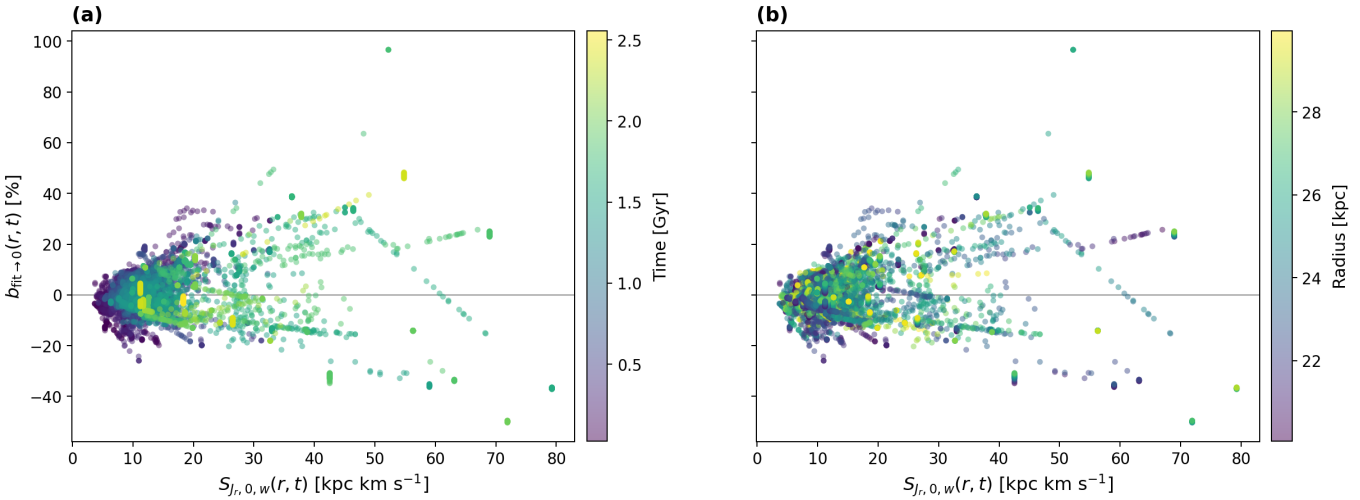


Figure 7. Relation between the weighted targeted radial-action spread, $S_{J_r,\mathrm{tar},w}$, and the signed fitted-to-targeted enclosed-mass bias, $b_{\mathrm{fit}\to\mathrm{tar}}$, for the 1:10 merger in the $20 \leq r/\mathrm{kpc} \leq 30$ radial range. The points are colored by snapshot time in panel **(a)** and by radius in panel **(b)**. The range of permitted signed mass biases widens as the target action cloud becomes more extended, but the bias can take either sign. Thus, broadening of the target action cloud is associated with reduced fitting reliability without uniquely determining the resulting mass error. Earlier snapshots also preferentially occupy the more extreme parts of the mass-bias envelope, although the mass consequence of over-clustering depends on the direction of the fitted displacement in NFW parameter space.

4.2.1. *Artificial over-clustering of a broadened target-action cloud*

The first failure mechanism arises when the selected cohort is not compact in radial action even in the target potential. We characterize this broadening using the weighted target-action spread $S_{J_r,\mathrm{tar},w}(r,t)$. We then compare it with the weighted KLD excess

$$\Delta D_{\mathrm{KL},w} = D_{\mathrm{KL},w}(\widehat{\boldsymbol{\theta}}) - D_{\mathrm{KL},w}(\boldsymbol{\theta}_{\mathrm{tar}}), \tag{64}$$

where $\boldsymbol{\theta}_{\rm tar}$ denotes the target-potential parameters and $\widehat{\boldsymbol{\theta}}$ denotes the KLD-best-fit parameters. Positive $\Delta D_{{\rm KL},w}$ means that the fitted potential produces a more strongly clustered action distribution than the target potential.

The current 1:10 result for $20 \leq r/{\rm kpc} \leq 30$ is shown in Figure 6. As $S_{J_r,{\rm tar},w}$ increases, the occupied region extends toward larger positive $\Delta D_{{\rm KL},w}$. The largest KLD excesses occur primarily among the more broadly distributed target action clouds, although the substantial scatter shows that target-action spread alone does not determine how strongly the fitted potential can overcluster the distribution.

The time coloring in Figure 6(a) adds a further distinction. Earlier snapshots preferentially populate the upper part of the distribution, while later snapshots with broad target-action clouds generally show smaller attainable KLD excesses. A simple physical interpretation is that a broad early-time action cloud can still arise from an orderly stream structure. Changing the trial potential can then move the assigned actions of its members coherently and artificially compress the cloud. After further merger evolution, the stellar kinematics contain more member-dependent disturbance. Changing a smooth spherical potential still remaps every assigned action, but it is less able to undo those unequal perturbations and produce a highly clustered distribution. This result indicates that high clustering is less accessible for those more irregular data within the tested NFW family; it does not imply a universal upper bound on KLD at late times.

The corresponding fitted-to-targeted mass bias is shown in Figure 7. The distribution develops a broader vertical envelope as $S_{J_r,{\rm tar},w}$ increases: compact target-action clouds are concentrated near small mass bias, whereas broad clouds permit both larger positive and larger negative biases. Earlier snapshots preferentially occupy the more extreme parts of this envelope, consistent with their greater susceptibility to artificial KLD compression. A large KLD excess gives the optimization a stronger preference for a potential away from the targeted solution, but it does not by itself determine how much the enclosed mass changes.

target-action broadening is therefore a risk indicator rather than a direct predictor of the sign or magnitude of the mass error. Its consequence depends both on how readily the broad cloud can be compressed by the trial-potential family and on the direction in parameter space selected by that compression. The latter dependence is examined in Section 4.2.2.

4.2.2. *KLD-score, action-value, and mass-direction degeneracies*

Over-clustering is not the only way for KLD maximization to fail. Even when the fitted and target action clouds have similar degrees of clustering, the KLD surface need not possess a unique maximum at the targeted parameters. We test this possibility by evaluating the radial actions and their KLD over a two-dimensional grid of trial NFW parameters,

$$\boldsymbol{\theta} = (\rho_0, r_s), \tag{65}$$

centered on the target potential. The same cohort and snapshot are used throughout each parameter grid.

Figure 8 shows the resulting KLD landscape for the selected cohort from parent stream 043 in the 1:10 merger at snapshot 55. The cohort has a median radius of $r_{\rm med} = 25.17\,{\rm kpc}$ and was selected from the $20 \leq r/{\rm kpc} \leq 30$ radial range. Rather than forming an isolated peak around the target potential, the largest KLD values occupy an elongated, approximately one-dimensional region in the (ρ_0, r_s) plane. The near-maximum contour consequently includes multiple combinations of density normalization and scale radius. The KLD score alone cannot strongly distinguish among trial potentials along this direction.

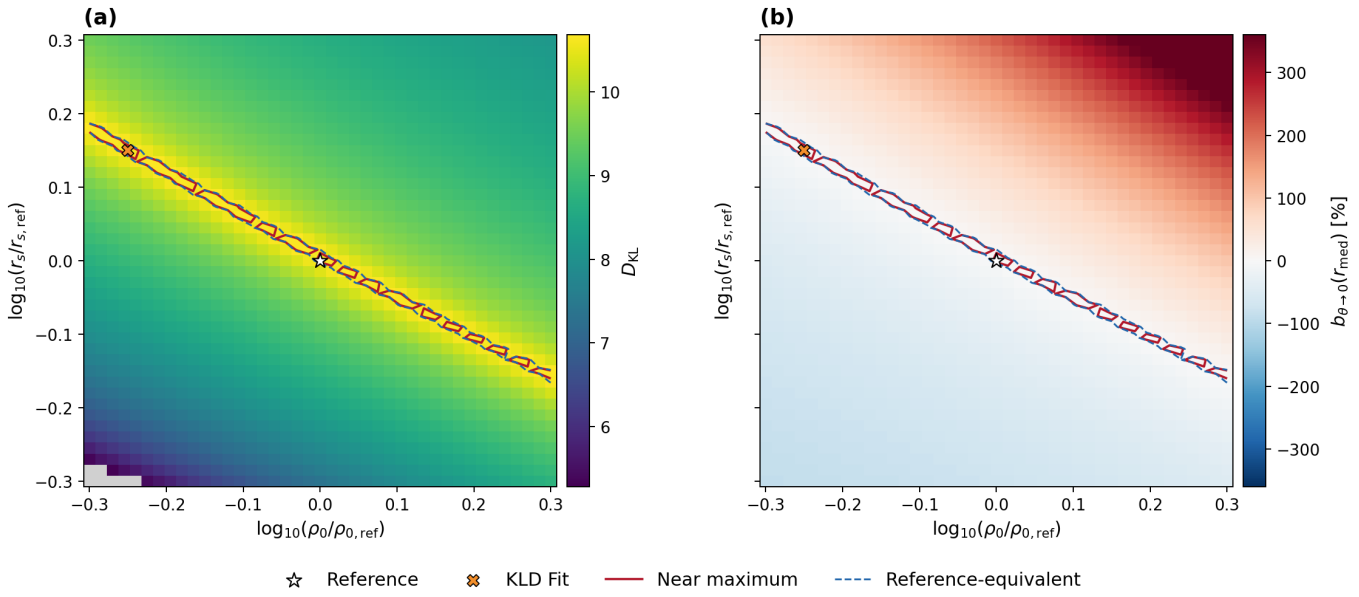


Figure 8. KLD-score degeneracy for a selected cohort from parent stream 043 in the 1:10 merger at snapshot 55. The cohort is selected from $20 \leq r/\mathrm{kpc} \leq 30$ and has median radius $r_{\mathrm{med}} = 25.17\,\mathrm{kpc}$. **(a)** Radial-action KLD evaluated over the trial NFW parameter grid. The axes give the logarithmic offsets of ρ_0 and r_s from their targeted values. The solid contour bounds the region whose KLD is indistinguishable from the grid maximum within the adopted finite-sample score tolerance. The dashed contour bounds trial potentials whose KLD is consistent with that of the targeted potential under the corresponding matched-star comparison. **(b)** Signed enclosed-mass difference between each trial potential and the target potential at r_{med}, $b_{\theta\to\mathrm{tar}}(r_{\mathrm{med}}) = 100[M_\theta(< r_{\mathrm{med}}) - M_{\mathrm{tar}}(< r_{\mathrm{med}})]/M_{\mathrm{true}}(< r_{\mathrm{med}})$. The target potential and the KLD fit are identified by the markers. The extended, nearly one-dimensional KLD contours demonstrate that similar clustering scores can be produced by different NFW parameter combinations. The KLD tolerances define numerical score equivalence and are not confidence intervals. Gray pixels indicate trial potentials for which the finite-action requirement is not satisfied. The mass effect of a parameter displacement depends on whether it follows or crosses the enclosed-mass contours in panel (b).

The companion mass map in panel (b) shows why an extended KLD ridge does not necessarily imply a large enclosed-mass bias. A change in (ρ_0, r_s) changes the inferred mass only to the extent that the parameter displacement crosses the enclosed-mass contours. A displacement along a nearly constant-mass direction can therefore produce substantially different NFW parameters while retaining a similar enclosed mass at the radius sampled by the cohort.

For the illustrated cohort, portions of the near-maximum and targeted-equivalent KLD regions follow directions that remain close to $b_{\theta\to\mathrm{tar}}(r_{\mathrm{med}}) = 0$. The KLD score is consequently insufficient to identify unique NFW parameters, but the locally enclosed mass can remain much better constrained than either parameter separately. Conversely, if a KLD-preferred displacement crosses the mass contours, the same parameter non-uniqueness can propagate into a substantial mass bias. The relevant issue is therefore not only the existence of a KLD excess or ridge, but also its orientation relative to the enclosed-mass map.

A second form of non-uniqueness concerns the characteristic action value itself. For every trial potential, we calculate

$$\Delta\,\mathrm{median}\,J_r(\boldsymbol{\theta}) = \mathrm{median}_a\,[J_{r,a}(\boldsymbol{\theta})] - \mathrm{median}_a\,[J_{r,a}(\boldsymbol{\theta}_{\mathrm{tar}})]\,, \tag{66}$$

where a indexes the stars used at that parameter-grid location. This quantity measures the displacement of the median radial action relative to the target potential; it does not measure the width or clustering of the action cloud.

Figure 9 shows that the region producing a targeted-like median action is also extended across the NFW parameter plane. Thus, substantially different combinations of ρ_0 and r_s can assign approximately the same median radial action to the selected cohort. The median radial action is therefore not an injective function of the NFW parameters.

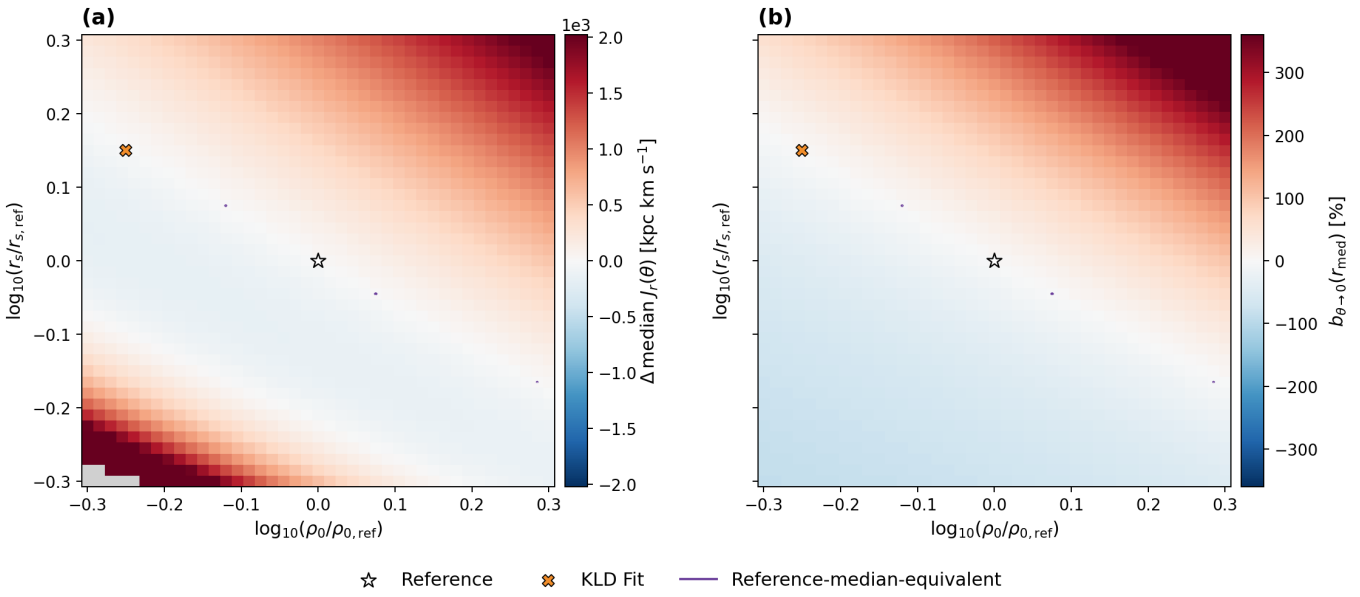


Figure 9. Median-action degeneracy for the same 1:10 cohort, snapshot, and radial range as Figure 8. **(a)** Signed difference between the median radial action in each trial potential and that in the target potential, $\Delta\,\mathrm{median}\,J_r(\boldsymbol{\theta})$. The contour marks $|\Delta\,\mathrm{median}\,J_r| = 0.10\,S_{J_r,\mathrm{tar}}$, where $S_{J_r,\mathrm{tar}}$ is the MAD width of the targeted radial-action cloud. It therefore identifies the boundary of the region whose median action is close to the targeted median under the adopted tolerance. **(b)** Signed enclosed-mass difference $b_{\theta\to\mathrm{tar}}(r_{\mathrm{med}})$ over the same parameter grid. The extended targeted-median-equivalent region demonstrates that recovery of the targeted median action does not uniquely recover the targeted NFW parameters. Gray pixels indicate trial potentials for which the finite-action requirement is not satisfied.

To control numerical truncation by stars with non-finite radial actions, we define a fixed core sample containing stars with finite actions over at least 75% of the trial-potential grid. A KLD value is retained at a given grid point only when at least 95% of this fixed core remains finite there. A separate map of the fraction of stars with non-finite actions is used to verify which parts of the parameter grid are affected by this truncation. This procedure keeps the stellar sample nearly fixed across the accepted grid while avoiding the removal of the entire map because of a small number of locally non-finite orbits.

The two figures diagnose distinct properties of the action cloud. Figure 8 concerns its degree of clustering, whereas Figure 9 concerns the location of its median radial action. A trial potential can therefore obtain a targeted-equivalent KLD score without reproducing the targeted median, or reproduce the targeted median without being uniquely identified by the KLD objective. Agreement in both scalar quantities would still not prove equality of the complete action distributions on a star-by-star basis.

These results establish the existence of KLD-score and median-action degeneracies for the illustrated cohort, but they also show that parameter identifiability and mass identifiability are different questions. A KLD-preferred displacement can be large in parameter space yet have little effect on the enclosed mass when it follows a nearly constant-mass direction. Only the component that crosses mass contours changes the inferred mass substantially. The present example demonstrates this directional dependence, but it does not determine how frequently favorable or unfavorable alignments occur across cohorts, snapshots, radial ranges, and merger histories. Over-clustering and degeneracy may also occur simultaneously; they are diagnostic failure routes, not additive components of the mass bias.

The directional result suggests an additional observationally available post-fit screen. For every selected cohort, one may identify the region in the parameter space occupied by the trial potentials whose KLD values are statistically or numerically indistinguishable from the maximum. Cohorts for which the near-maximum region spans a narrow mass range may be retained or given greater weight, whereas cohorts permitting a broad mass range should be downweighted or rejected. This criterion preserves cases in which the NFW parameters are non-unique but the local enclosed mass is identifiable. In the NFW examples examined here, adding this preference is compatible with the original cohort selection; a systematic test over a broader potential family remains future work. This additional screen tests mass identifiability rather than mass accuracy, because an artificially over-clustered near-maximum region can still be displaced as a whole from the target potential.

4.3. *An exploratory bending constraint*

Since KLD alone can favor several different potential models, the cohort geometry may provide another way to compare them. However, a stream's present shape records its past evolution, so an instantaneous potential alone does not determine its current curvature. Although a candidate potential gives the acceleration of a star at a known position, direct acceleration measurements are not part of the stream data assumed here. We therefore restrict the comparison to cohorts whose local motions suggest that their tracks may approximate stellar trajectory segments.

To describe the measured track in the adopted projection, let s be the distance along it and $\psi(s)$ the direction of its tangent relative to a fixed axis. Its signed curvature, $\kappa = d\psi/ds$, measures how much the track turns per unit distance, with a larger absolute value indicating a sharper bend. We orient the normal $\hat{\boldsymbol{n}}$ to the left of the tangent, so positive and negative curvature distinguish the two turning directions. For a star currently on the track, the normal component of its projected acceleration satisfies

$$a_n = v_\parallel^2 \kappa + \dot{v}_n. \tag{67}$$

Here $v_\parallel$ describes motion along the track, whereas v_n describes sideways motion relative to it. The first term is the normal acceleration needed for a star to follow the track's bend at speed $|v_\parallel|$; the remaining term, $\dot{v}_n$, describes the change in its sideways velocity relative to the fixed present-day track. Thus, a small v_n at one instant does not establish a small $\dot{v}_n$, since a star can be aligned with the track while beginning to depart from it.

If the track locally represents the stellar trajectory and $\dot{v}_n$ is negligible, the model acceleration can be used to infer the turn along a segment. For a candidate potential Φ_θ, with $\boldsymbol{a}_\theta = -\nabla\Phi_\theta$, we define

$$B_{\theta,j} = \int_{s_{j,1}}^{s_{j,2}} \frac{\boldsymbol{a}_{\theta,\mathrm{proj}} \cdot \hat{\boldsymbol{n}}}{v_\parallel^2}\, ds, \tag{68}$$

and compare it with the measured turn,

$$B_{\mathrm{obs},j} = \psi(s_{j,2}) - \psi(s_{j,1}). \tag{69}$$

The measured quantity is the change in direction between the two ends of the present track, while the model quantity is the turn implied by the candidate force under the local-orbit approximation. A stronger normal force produces more turning over a given length, whereas a larger speed allows less turning over the same length. Since both quantities use the same track, projection, interval, and sign convention, they can be compared without generating a stream model or reconstructing its formation history.

To identify plausible cases for this approximation, we retain the original cohort members and add two velocity cuts. In populated bins along the track, we require the member-weighted RMS angle between the mean three-dimensional velocity and the fitted three-dimensional tangent to be at most 5°, using one flow direction for the entire cohort. This checks whether the mean stellar motion follows the spatial track. We also require the member-weighted RMS of $\sigma_{v,j}/|\overline{\boldsymbol{v}}_j|$ to be at most 0.05, where $\sigma_{v,j}$ is the three-dimensional velocity scatter after removing a local linear trend. This ratio measures the random velocity scatter relative to the local streaming speed; removing the trend allows the mean velocity to change smoothly along a curved cohort. We require at least three bins with at least ten stars each so that both cuts can be estimated over several populated parts of the track.

Because nearly straight or poorly resolved tracks give unstable bend comparisons, we also require a supported bend of at least 15°, a bootstrap signal-to-scatter ratio of at least three, and a change of at most 25% under the broad-binning check. The bootstrap ratio compares the measured bend with its scatter across 200 resampled stellar sets, while the binning check tests whether the bend depends strongly on the chosen broad bins. We measure the bend in the x–y projection and restrict its endpoints to the populated track. Each comparison window must retain a bend of at least 5°, a signal-to-scatter ratio of at least three, a projected velocity–tangent angle below 15°, and $|v_\parallel| \geq 10\,\mathrm{km\,s^{-1}}$ throughout the interval. These last conditions avoid dividing by an unresolved bend or a nearly zero along-track speed, but none of the cuts measures $\dot{v}_n$.

For the usable windows $\mathcal{W}$, we combine the fractional bend differences into

$$D_B(\theta) = \left[\frac{1}{|\mathcal{W}|}\sum_{j\in\mathcal{W}}\left(\frac{B_{\theta,j} - B_{\mathrm{obs},j}}{B_{\mathrm{obs},j}}\right)^2\right]^{1/2}. \tag{70}$$

Dividing by the measured bend expresses each mismatch relative to the turn being tested, so $D_B = 0$ means agreement in every window and $D_B = 0.2$ corresponds to an RMS fractional mismatch of 20%. Since the bends retain their signs, a model that turns in the opposite direction is penalized even when its bend magnitude is correct. The windows overlap, however, so D_B is a descriptive discrepancy rather than an independent-window likelihood.

To preserve the information from action clustering, we minimize D_B only among candidates whose KLD scores remain close to the maximum. Writing $K(\theta)$ for the original KLD score, we select

$$\widehat{\theta}_\delta = \underset{\substack{\theta\in\Theta_{\mathrm{sampled}}\\ K(\theta)\geq K_{\max}-\delta}}{\arg\min}\; D_B(\theta). \tag{71}$$

Here $\Theta_{\mathrm{sampled}}$ is the set of evaluated models within the original fitting bounds, and δ is the largest KLD decrease allowed relative to its maximum. Thus, a larger δ

gives the bend comparison more freedom to move away from the KLD optimum; it is an absolute score tolerance, not a confidence interval or a percentage loss. We retain the original maximum-KLD model as the $\delta = 0$ control and test $\delta = 0.01, 0.03, 0.1, 0.2, 0.3$, keeping the original action estimator and finite-action treatment. The target is not used to choose a tolerance.

At snapshot 55, three of the 106 original cohort records pass all checks: cohort 213 at 20–30 kpc in the 1:5 and 1:10 mergers, and cohort 836 at 50–70 kpc in the 1:100 merger. To assess whether the additional constraint helps, we freeze the model choices before reading the stored target parameters and measure their separation using

$$d_\theta = \left[\ln^2\left(\frac{\rho_0}{\rho_{0,\mathrm{tar}}}\right) + \ln^2\left(\frac{r_s}{r_{s,\mathrm{tar}}}\right)\right]^{1/2}. \tag{72}$$

This distance is zero at the target, and the logarithms compare multiplicative departures without mixing the units of density and length. For example, if one parameter is exact while the other is wrong by a factor of two, then $d_\theta = \ln 2$, regardless of which parameter is wrong. A negative change in d_θ relative to the KLD-only result therefore indicates improved parameter recovery, although the distance is not a parameter uncertainty. For every tested positive tolerance, d_θ decreases for the 1:5 and 1:100 cohorts but increases for the 1:10 cohort, whose KLD-only model is already close to its target. Consequently, a smaller bend discrepancy does not always give more accurate parameters. No simulation stellar acceleration enters the selection or bend comparison.

Figure 10 illustrates the 1:100 result. Although its KLD-only model has a lower ρ_0 than the target, its larger r_s gives a greater enclosed mass at the cohort's median radius. Since both parameters enter the NFW mass profile, their combined effect determines the mass rather than the density normalization alone. Moreover, KLD rewards concentration in the assigned actions, so an incorrect potential can obtain a higher score by compressing that distribution, as discussed in Section 4.2. The map is consistent with this failure route, although it does not establish artificial compression as the sole cause of the KLD excess in this cohort.

The right panel compares each model's enclosed mass with that of the target at the median member radius r_{med}. Its signed value $b_{\theta\to\mathrm{tar}}$ is positive when the model mass is larger and negative when it is smaller, with the true enclosed mass used only for normalization. Although the refined 1:100 models lie closer to the target parameters, they remain on the positive side of this map. We therefore report the changes in parameter distance and absolute mass discrepancy separately in the caption, since improvement in one does not imply improvement in the other. Taken together, the three survivors show that geometry can help individual fits, but they do not establish a general correction for KLD fitting failures.

5. SIMULATION-ONLY DIAGNOSTICS OF THE FULL POTENTIAL

The target-potential and fitted-to-targeted diagnostics in Section 4 require knowledge of the simulated force field and cannot be reproduced directly from Milky Way observations. We now introduce a second simulation-only comparison that tests the radial-orbit mapping of the fitted spherical model against that of the frozen full BFE field.

5.1. *Frozen-snapshot Radial-cycle Diagnostic*

At a snapshot t_*, we freeze the BFE potential and integrate a model orbit from each measured stellar phase-space point. If $\tau_{p,k}$ and $\tau_{p,k+1}$ are successive pericentric passages of that frozen orbit, we define

$$\widetilde{J}_{r,\mathrm{BFE}}(t_*) \equiv \frac{1}{2\pi}\int_{\tau_{p,k}}^{\tau_{p,k+1}} p_r(\tau)\,\dot{r}(\tau)\,d\tau = \frac{1}{2\pi}\int_{\text{radial cycle}} p_r\,dr. \tag{73}$$

In an integrable spherical potential this construction reduces to the canonical radial action. In the frozen, non-spherical merger potential, however, one radial oscillation need not be a fundamental cycle of an invariant three-torus. The result may depend on phase, cycle number, and the radial-cycle definition. We therefore treat $\widetilde{J}_{r,\mathrm{BFE}}$ as a radial-cycle diagnostic rather than a canonical action.

For cohort i, using only stars with finite values in both calculations, we define

$$\Delta\widetilde{J}_{r,\mathrm{med},i} = \mathrm{median}(J_{r,\mathrm{fit},i}) - \mathrm{median}(\widetilde{J}_{r,\mathrm{BFE},i}) \tag{74}$$

and

$$\Delta D_{\mathrm{KL},r,i} = D_{\mathrm{KL}}(J_{r,\mathrm{fit},i}) - D_{\mathrm{KL}}(\widetilde{J}_{r,\mathrm{BFE},i}). \tag{75}$$

The matched-star-count-weighted diagnostics are

$$\left\langle \Delta\widetilde{J}_{r,\mathrm{med}} \right\rangle_w = \frac{\sum_i N_{i,\mathrm{match}}\Delta\widetilde{J}_{r,\mathrm{med},i}}{\sum_i N_{i,\mathrm{match}}}, \tag{76}$$

$$\langle \Delta D_{\mathrm{KL},r} \rangle_w = \frac{\sum_i N_{i,\mathrm{match}}\Delta D_{\mathrm{KL},r,i}}{\sum_i N_{i,\mathrm{match}}}. \tag{77}$$

These quantities do not supply the theoretical basis of the KLD fit and are not observable inputs. They diagnose disagreement between the radial-orbit structures generated by the fitted spherical model and by the frozen full simulated field.

5.2. *Outer-halo Radial-orbit Disagreement*

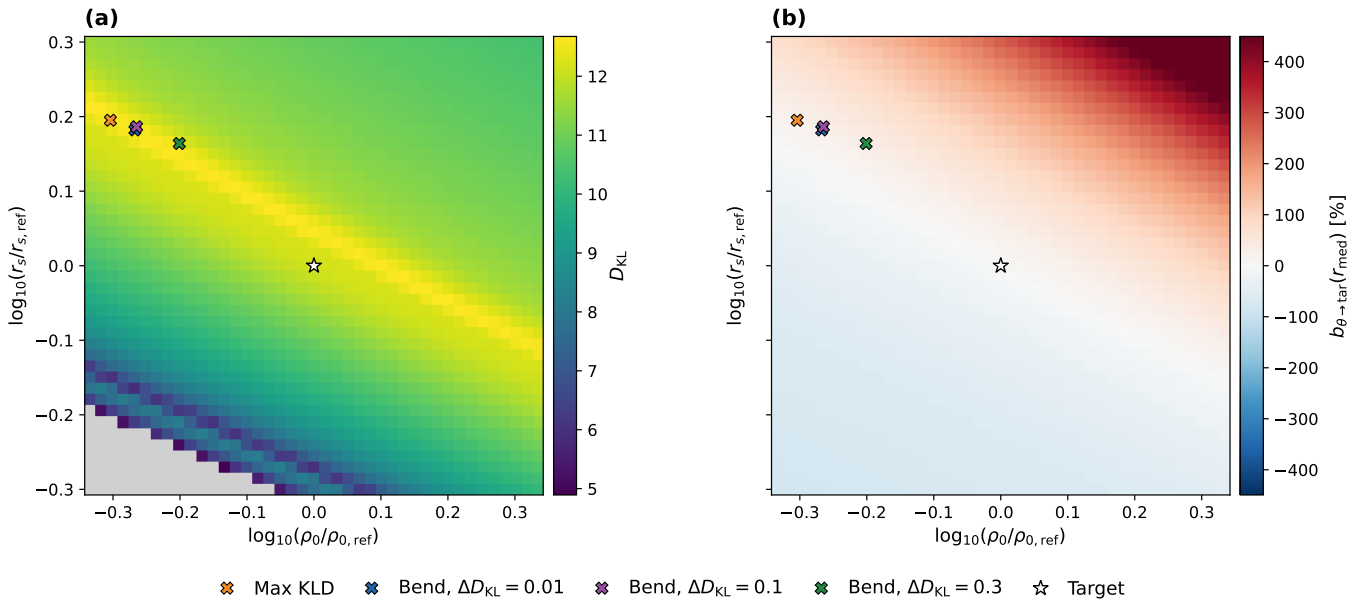


Figure 10. Bending refinement for cohort 836 in the 1:100 merger at snapshot 55, selected from 50–70 kpc. (a) The radial-action KLD over the trial NFW parameter grid. (b) The signed enclosed-mass difference from the target, $b_{\theta\to\mathrm{tar}}(r_{\mathrm{med}}) = 100[M_\theta(< r_{\mathrm{med}}) - M_{\mathrm{tar}}(< r_{\mathrm{med}})]/M_{\mathrm{true}}(< r_{\mathrm{med}})$. The orange cross marks Max KLD, the other crosses show the three bend-refined choices, and the white star marks Target. Original fitting bounds are $5 \times 10^5 \leq \rho_0/(\mathrm{M}_\odot\,\mathrm{kpc}^{-3}) \leq 10^7$ and $5 \leq r_s/\mathrm{kpc} \leq 30$. The display grid can extend beyond these bounds, but the five marked models are the frozen choices from the original fit. Gray cells indicate an undefined KLD. Mass colors cover the full grid and saturate at a symmetric limit set by its 98th percentile in absolute value. The refined models lie closer to Target in parameter space, but remain on the positive side of the mass-difference map.

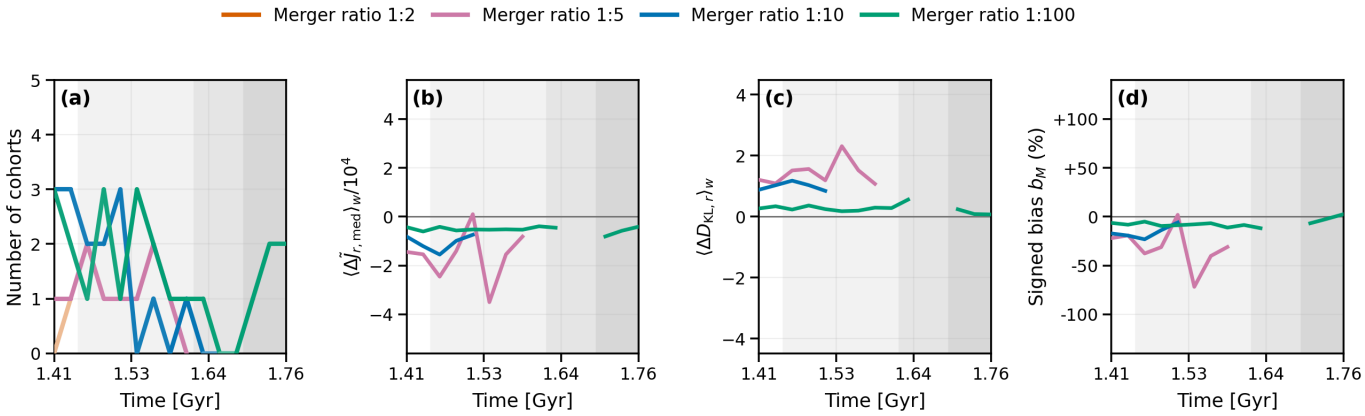


Figure 11. Comparison of the KLD-fitted spherical NFW model with the frozen-snapshot BFE field for merger mass ratios 1:2, 1:5, 1:10, and 1:100 in the $80 \leq r/\mathrm{kpc} \leq 100$ radial range. Only BFE-valid snapshot intervals within $1.41 \leq t/\mathrm{Gyr} \leq 1.76$ contribute. **(a)** Number of selected coherent cohorts. Color identifies the merger ratio, while the saturation of each curve segment encodes the average stellar membership of the selected cohorts. More saturated segments correspond to larger average membership. **(b)** Matched-star-count-weighted signed median displacement, $\langle \Delta \tilde{J}_{r,\mathrm{med}} \rangle_w$, scaled by 10^4 as indicated on the axis. **(c)** Matched-star-count-weighted signed clustering difference, $\langle \Delta D_{\mathrm{KL},r} \rangle_w$. **(d)** Signed enclosed-mass bias, b_M, of the fitted spherical model relative to the globally spherically averaged BFE mass profile. The action diagnostics are defined in Equations 76 and 77. White background denotes times at which all four merger cases lie within BFE-valid intervals; progressively darker gray denotes coverage by fewer cases. Missing segments occur when the BFE comparison is unavailable or when no selected cohort contributes a finite measurement.

Figure 11 compares the evolution of the selected stream sample, the disagreement between radial actions in the KLD-fitted spherical potential and radial-cycle integrations in the frozen BFE field, and the resulting

enclosed-mass bias in the outer-halo range $80 \leq r/\mathrm{kpc} \leq 100$. The action differences and the fitted mass bias exhibit related time dependence. This covariation indicates that disagreement in the radial orbital structure generated by the fitted spherical and frozen full potentials accompanies poor mass recovery. It does not, by itself, establish that the radial-action difference is an independent or additive source of the mass bias. Rather, it provides a dynamical diagnostic of the mismatch already present between the fitted and true orbital mappings.

The figure is restricted to $1.41 \leq t/\mathrm{Gyr} \leq 1.76$, which provides the largest overlapping BFE-valid coverage across the merger cases. This window does not include the principal BFE-valid interval of every individual simulation. In particular, much of the BFE-valid evolution of the 1:2 merger lies outside the displayed interval, so that case contributes only limited information to the comparison. The apparent differences among merger ratios in this restricted window should therefore be interpreted together with their unequal temporal coverage.

5.3. *Dependence on Perturber Orbital Configuration*

Figure 12 compares the standard, circular, and radial 1:5 merger configurations over their respective BFE-valid intervals. The radial configuration appears most favorable for the KLD fitting method over the periods analyzed here. Although the close radial encounter may produce a strong transient disturbance, the strongest forcing is plausibly concentrated into a relatively short interval. Outside this interval, coherent cohorts remain or reappear, and the fitted-minus-BFE action discrepancies and enclosed-mass biases are comparatively modest over much of the sampled evolution.

The circular configuration is less favorable in terms of sustained tracer availability. One possible interpretation is that prolonged merger-induced forcing continuously distorts the halo and stream population. The surviving action clouds generally have positive $\Delta D_{\mathrm{KL},r}$ and negative $\Delta \widetilde{J}_{r,\mathrm{med}}$, indicating that the optimized spherical NFW model produces an action distribution that is more clustered and shifted toward lower radial actions relative to the BFE field.

This orbital comparison should be regarded as suggestive rather than as a universal ranking of perturber trajectories, since the analysis is restricted to intervals in which the BFE provides a sufficiently reliable representation. However, the comparison indicates that not only merger mass ratio, but also the duration and geometry of the encounter, affect the survival of coherent stream material and the form in which the action-space fitting failure appears.

6. DISCUSSION AND CONCLUSIONS

6.1. *What Observable Cohort Selection Can and Cannot Guarantee*

The selection procedure developed in Sections 2.5 and 3.2 removes stream material that is visibly folded, phase mixed, or kinematically irregular. A surviving cohort is therefore more likely to have experienced an approximately common target-action change, leaving a compact cloud that can support action-clustering inference. This is the principal reason that the post-selection sample can recover a useful inner-halo mass profile at the distance-matched 1:10 snapshot.

The selection remains a probabilistic dynamical indicator rather than a proof. A cohort may pass the present-day cuts even if its target-action cloud was broadened to some extent by earlier member-dependent forcing. Conversely, the absence of a cohort at one snapshot does not imply that every material stream has been destroyed; it means that no tested segment then satisfies the adopted observable requirements. Because selection is repeated independently, the cohorts contributing at different snapshots need not be the same.

The remaining failure routes occur at three levels. First, the system may no longer contain enough selected cohorts to constrain the potential. Second, surviving cohorts may fail to recover their spherical target potentials because a different trial potential artificially overclusters the assigned actions or because the KLD surface is degenerate. Third, even exact target-potential recovery need not reproduce the globally averaged true mass profile. The exact fitted-to-targeted plus targeted-to-true decomposition prevents these logically distinct effects from being conflated. In the illustrated 1:10 inner-halo case, the target-to-true contribution is generally smaller and more spatially restricted, whereas failure of the KLD fit to recover the targeted spherical model supplies most of the extended mass-bias structure.

The parameter-plane diagnostics add a separate directional condition. KLD overestimation or parameter nonuniqueness changes the inferred mass only if the target-to-fitted displacement in the parameter space crosses the enclosed-mass contours. A displacement along a nearly constant-mass direction can leave the locally inferred mass nearly unchanged even when the KLD or NFW parameters differ from their target values. The post-fit screen proposed in Section 4.2.2 therefore evaluates the enclosed-mass range of the full near-maximum KLD region rather than requiring a unique parameter maximum.

The curvature constraint of J. Nibauer et al. (2023) motivates using stream geometry, but its directional information cannot distinguish the spherical NFW models fitted here. For a fixed center, all these models accelerate stars toward the center, regardless of their density or scale radius. Changing the parameters therefore changes the force strength while leaving its direction unchanged.

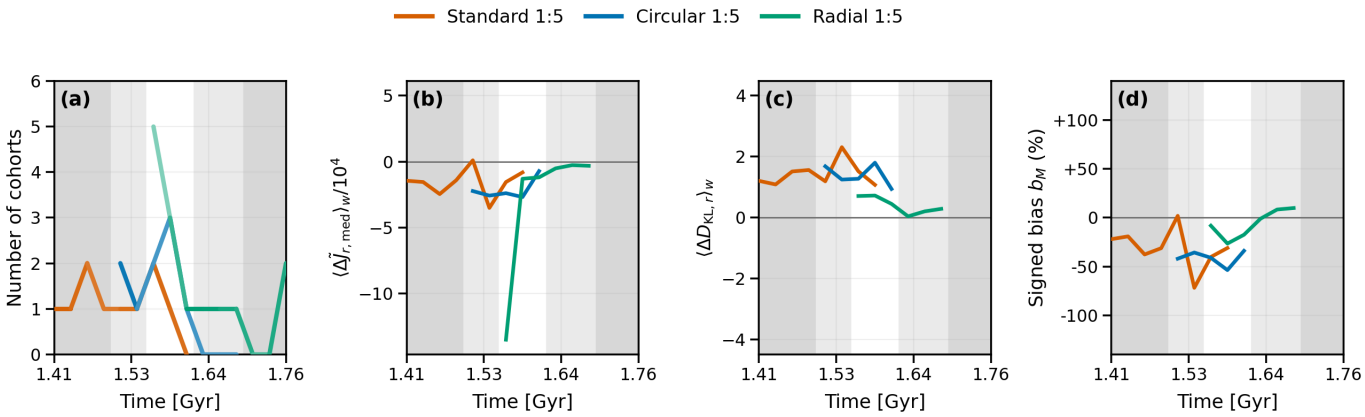


Figure 12. Comparison of the KLD-fitted spherical NFW model with the frozen-snapshot BFE field for the standard, circular, and radial 1:5 merger configurations in the $80 \leq r/\mathrm{kpc} \leq 100$ radial range. Only BFE-valid snapshot intervals within $1.41 \leq t/\mathrm{Gyr} \leq 1.76$ contribute to the comparison. **(a)** Number of selected coherent cohorts. Curve color identifies the orbital configuration, while increasing color saturation represents increasing average stellar membership per cohort. **(b)** Matched-star-count-weighted signed median displacement, $\langle \Delta \widetilde{J}_{r,\mathrm{med}} \rangle_w$, scaled by 10^4 as indicated on the axis. **(c)** Matched-star-count-weighted signed clustering difference, $\langle \Delta D_{\mathrm{KL},r} \rangle_w$. **(d)** Signed enclosed-mass bias, b_M, of the fitted spherical model relative to the globally spherically averaged BFE mass profile. White background indicates times at which all three configurations lie within BFE-valid intervals; progressively darker gray indicates that fewer configurations provide a valid BFE comparison. Missing curve segments indicate that no valid diagnostic is available.

Since the directional test cannot separate these models, we instead ask whether the measured amount of bending can help distinguish candidates with similarly large KLD scores.

This comparison requires a link between force strength and track curvature. If the track locally approximates a stellar trajectory, its curvature is approximately the normal acceleration divided by the squared along-track speed. Thus, at a fixed speed, a stronger normal force implies a sharper turn. The velocity cuts in Section 4.3 identify cohorts for which this approximation may be useful, while the snapshot-55 pilot tests whether it actually improves parameter recovery.

Although the additional velocity cuts favor cohorts that currently move along their tracks, they cannot establish that this alignment persists. In particular, a star with small v_n can still have a large $\dot{v}_n$, which describes its developing sideways departure from the present track. Since earlier forcing and changes in the stream shape are not recovered from these instantaneous cuts, a cohort may pass them while still failing the local-orbit approximation needed for the bend comparison.

In the pilot, two survivors improve in parameter distance, whereas the third worsens. The current evidence therefore supports an exploratory use of geometry for individual cohorts, while a general correction or weighting rule would require broader validation.

Several validation quantities used in this work require simulation knowledge: the true time-dependent force, the cohort-wise target-potential sequence, target actions, the direct dark-matter enclosed mass, and the frozen-BFE radial-cycle integrations. In Galactic data, reliability must instead be assessed through observable coherence, resampling stability, agreement among independently fitted cohorts, consistency across radial ranges or action components, sensitivity to fitting bounds, and the mass range spanned by the near-maximum KLD region.

6.2. *Future Potential Families*

The spherical NFW family is deliberately economical and keeps the actions well defined and computationally tractable, but it cannot represent the full angular structure of a merger-distorted halo. A natural next step is to repeat the analysis with an axisymmetric or triaxial action-computable family. Such a model could reduce the target-to-true mismatch and test whether some fitted-to-target failures currently attributed to over-clustering instead reflect insufficient flexibility of the spherical family. The extension must preserve a well-defined action construction—for example through an integrable or controlled approximately integrable model—because a generic triaxial potential is not automatically integrable. It will also require more stellar coverage and stronger regularization to prevent the added parameters from worsening the KLD degeneracies diagnosed here.

For the fixed-center spherical family used here,

$$\boldsymbol{a}_\theta(\boldsymbol{r}) = -\frac{GM_\theta(<r)}{r^2}\hat{\boldsymbol{r}}. \tag{78}$$

Since this force always points toward the center, changing ρ_0 or r_s changes its magnitude but not its direction. A direction-only test therefore cannot distinguish the spherical models considered here. For a future triaxial model, however, parameters that alter the gravitational

force direction could be checked against the stream geometry, provided the coherence condition remains appropriate.

6.3. *Conclusions*

Our principal conclusions are as follows:

1. A time-dependent non-integrable merger field can be analyzed through cohort-specific sequences of instantaneously frozen spherical target potentials without treating the true potential as integrable.

2. The assigned target action evolves through an exactly derived residual-force term and a target-map evolution term. Observable cohort selection is designed to favor material for which the member-to-member differences in their accumulated sum of those two terms remain small.

3. Selected cohorts recover a useful inner-halo enclosed-mass profile at the distance-matched 1:10 snapshot, but stronger mergers generally shorten the interval of stable recovery and eventually reduce the supply of usable material.

4. Broadened target-action clouds are more vulnerable to artificial KLD compression, although at later merger stages, more irregular kinematics can be less readily mapped into a highly clustered cloud. The spread of target-action is therefore a risk indicator, not a deterministic predictor of the magnitude or the sign of the mass-bias.

5. The degeneracies of the KLD-score and the median-action need not imply a large mass error. Their consequence depends on the orientation from the target potential to the KLD-preferred models in the parameter space relative to the enclosed-mass contours, motivating a post-fit mass-identifiability screen.

6. With stricter velocity cuts, a bending comparison improves target-parameter recovery for two of three surviving cohorts at snapshot 55 and worsens it for the third. This pilot shows a possible use of geometry, but also the limits of inferring trajectory curvature from a present-day stream track.

7. Observable selection improves robustness but cannot guarantee unique potential recovery or an unbiased global mass profile. Simulation-only BFE comparisons remain valuable for calibrating these limitations and for guiding future applications to Galactic data.

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